

Package ‘Rsurrogate’

August 19, 2026

Type Package

Title Robust Estimation of the Proportion of Treatment Effect
Explained by Surrogate Marker Information

Version 3.3

Maintainer Layla Parast <parast@austin.utexas.edu>

Description

Provides functions to estimate the proportion of treatment effect on the primary outcome that is explained by the treatment effect on the surrogate marker. A tutorial for this package can be found at <<https://github.com/laylaparast/Rsurrogate>> and a Shiny App implementing the package can be found at <<https://parastlab.shinyapps.io/RSurrogateApp/>>.

License GPL

Imports stats, survival, Matrix, glmnet

NeedsCompilation no

Author Layla Parast [aut, cre]

Depends R (>= 3.5.0)

Contents

.bootstrap_var	2
.debias_deltas	3
.debias_ols	4
.debias_p	5
.debias_pred	5
.full.surrogate.approach	6
.make.delta.s	7
.smoothed.regularized	8
.tar.approach	9
.te.calculate	10
Aug.R.s.surv.estimate	11
augment.est.vector	15
calculate.var.np	15
cursor.weight	16
cumsum2	17
delper.si	18
delta.estimate	18
delta.multiple.surv	20

delta.s.estimate	22
delta.s.surv.estimate	23
delta.surv.estimate	25
delta.t.surv.estimate	27
d_example	28
d_example_me	29
d_example_multiple	30
d_example_surv	30
d_tar_example	31
fieller.calculate.me	32
fieller.ci	33
Kern.FUN	34
me.variance.estimate	34
perturb.nu.vector	35
pred.smooth	35
pred.smooth.surv	36
R.multiple.surv	37
R.s.estimate	42
R.s.estimate.me	46
R.s.surv.estimate	53
R.t.surv.estimate	57
select.surrogate	60
VTM	62
Index	64

.bootstrap_var	<i>Bootstrap variance estimation for select.surrogate</i>
----------------	---

Description

Helper function used by select.surrogate when var = TRUE; this resamples both treatment groups with replacement, reruns the entire estimation procedure on each resample, and summarizes the resulting bootstrap distribution of delta, delta.s, and the proportion of treatment effect explained.

Usage

```
.bootstrap_var(data, fit_once, boot.num)
```

Arguments

data	data frame of all observations (treated and control), including columns y, treat, and one column per candidate marker.
fit_once	a function of a single data frame argument (of the same form as data) that runs the point estimation procedure once and returns a list with elements delta, delta.s, and pte.
boot.num	number of bootstrap resamples.

Value

A list is returned:

<code>delta_se</code>	bootstrap standard error of delta.
<code>delta_ci</code>	a vector of size 2; the 2.5th and 97.5th percentiles of the bootstrap replicates of delta.
<code>delta.s_se</code>	bootstrap standard error of delta.s.
<code>delta.s_ci</code>	a vector of size 2; the 2.5th and 97.5th percentiles of the bootstrap replicates of delta.s.
<code>R.s_se</code>	bootstrap standard error of the proportion of treatment effect explained.
<code>R.s_ci</code>	a vector of size 2; the 2.5th and 97.5th percentiles of the bootstrap replicates of the proportion of treatment effect explained.
<code>boot.num</code>	the number of bootstrap resamples used.

Author(s)

Layla Parast

References

Parast, L., Freestone, G., Tarr, J. (2026). Treatment-Aware Regularization via Multi-Objective Optimization for Surrogate Selection.

<code>.debias_deltas</code>	<i>Residual treatment effect from a debiased refit, for one cross-validation fold</i>
-----------------------------	---

Description

Helper function used by `select.surrogate` with `selection = "TAR"`; this computes the residual treatment effect, evaluated on a held-out cross-validation fold, implied by a debiased coefficient vector, used to check whether a given penalty value keeps the estimated proportion of treatment effect explained within the tolerance epsilon of the full candidate set.

Usage

```
.debias_deltas(debias_coef, data_val, control_data)
```

Arguments

<code>debias_coef</code>	a one-column matrix of debiased coefficients, as returned by <code>.debias_ols</code> , with row names "Intercept" followed by the names of the selected markers.
<code>data_val</code>	data frame of held-out (validation fold) treated observations, including a column <code>y</code> and one column for each marker named in <code>debias_coef</code> .
<code>control_data</code>	data frame of control observations, including a column <code>y</code> and one column for each marker named in <code>debias_coef</code> .

Value

a numeric value; the residual treatment effect for the selected markers, evaluated on data_val and control_data, or NA if debias_coef contains missing values (an unidentified debiased refit).

Author(s)

Layla Parast

References

Parast, L., Freestone, G., Tarr, J. (2026). Treatment-Aware Regularization via Multi-Objective Optimization for Surrogate Selection.

.debias_ols	<i>Debiased ordinary least squares refit on lasso-selected markers</i>
-------------	--

Description

Helper function used by select.surrogate with selection = "TAR"; given a vector of penalized lasso coefficients, this refits an unpenalized (ordinary least squares) model on the treated group using only the markers with nonzero coefficients, to remove the shrinkage bias of the lasso.

Usage

```
.debias_ols(original_coefs, data_train)
```

Arguments

- original_coefs a named numeric vector of penalized lasso coefficients (excluding the intercept), for one value of the penalty.
- data_train data frame of treated observations, including a column y (the outcome) and one column for each marker named in original_coefs.

Value

a one-column matrix of the debiased (unpenalized) coefficients, with row names "Intercept" followed by the names of the selected markers. If the number of selected markers is at least as large as the number of treated observations, so that the debiased refit is not identified, a matrix of the same shape with all entries set to NA is returned instead.

Author(s)

Layla Parast

References

Parast, L., Freestone, G., Tarr, J. (2026). Treatment-Aware Regularization via Multi-Objective Optimization for Surrogate Selection.

<code>.debias_p</code>	<i>Counts the number of selected markers</i>
------------------------	--

Description

Helper function used by `select.surrogate` with `selection = "TAR"`; this counts the number of nonzero (non-intercept) coefficients in a debiased coefficient vector, used to track the size of the selected set along the weighted lasso path.

Usage

```
.debias_p(debias_coef, penalty)
```

Arguments

<code>debias_coef</code>	a one-column matrix of debiased coefficients, as returned by <code>.debias_ols</code> , with row names "Intercept" followed by the names of the selected markers.
<code>penalty</code>	not used; retained for compatibility when this function is applied across a list of debiased coefficient matrices, one per penalty value.

Value

an integer; the number of selected (non-intercept, nonzero) markers.

Author(s)

Layla Parast

References

Parast, L., Freestone, G., Tarr, J. (2026). Treatment-Aware Regularization via Multi-Objective Optimization for Surrogate Selection.

<code>.debias_pred</code>	<i>Predicted outcome from a debiased refit</i>
---------------------------	--

Description

Helper function used by `select.surrogate` with `selection = "TAR"`; this computes fitted values on a validation set from the debiased coefficients returned by `.debias_ols`, and is used to evaluate the cross-validated mean squared error along the weighted lasso path.

Usage

```
.debias_pred(debias_coef, data_val)
```

Arguments

<code>debias_coef</code>	a one-column matrix of debiased coefficients, as returned by <code>.debias_ols</code> , with row names "Intercept" followed by the names of the selected markers.
<code>data_val</code>	a matrix or data frame of held-out (validation fold) observations, including a column named Intercept (all ones) and one column for each marker named in <code>debias_coef</code> .

Value

a numeric vector of predicted outcome values for the observations in `data_val`.

Author(s)

Layla Parast

References

Parast, L., Freestone, G., Tarr, J. (2026). Treatment-Aware Regularization via Multi-Objective Optimization for Surrogate Selection.

`.full.surrogate.approach`

Estimates the proportion of treatment effect explained using all candidate markers

Description

Helper function used by `select.surrogate` with `selection = "none"`; this is a thin wrapper around `R.s.estimate`, run on all candidate markers, that adds a bootstrap variance option in place of `R.s.estimate`'s perturbation-resampling.

Usage

```
.full.surrogate.approach(data, surrogate_names, var = FALSE, boot.num = 200)
```

Arguments

<code>data</code>	data frame of all observations (treated and control), including columns <code>y</code> , <code>treat</code> , and one column per candidate marker.
<code>surrogate_names</code>	character vector of the candidate marker column names in <code>data</code> .
<code>var</code>	TRUE or FALSE; indicates whether a bootstrap variance estimate is requested, default is FALSE.
<code>boot.num</code>	number of bootstrap resamples used when <code>var = TRUE</code> , default is 200.

Value

A list is returned:

<code>delta</code>	the estimated overall treatment effect.
<code>delta.s</code>	the estimated residual treatment effect using all candidate markers.
<code>pte</code>	the estimated proportion of treatment effect explained using all candidate markers.
<code>var</code>	a list, as returned by <code>.bootstrap_var</code> , of bootstrap standard errors and confidence intervals for <code>delta</code> , <code>delta.s</code> , and <code>pte</code> ; NULL if <code>var = FALSE</code> .

Author(s)

Layla Parast

References

- Parast, L., Freestone, G., Tarr, J. (2026). Treatment-Aware Regularization via Multi-Objective Optimization for Surrogate Selection.
- Parast, L., McDermott, M., Tian, L. (2016). Robust estimation of the proportion of treatment effect explained by surrogate marker information. *Statistics in Medicine*, 35(10):1637-1653.

<code>.make.delta.s</code>	<i>Calculates the residual treatment effect for a selected set of markers</i>
----------------------------	---

Description

Helper function used by `select.surrogate` with `selection = "TAR"` or `selection = "rr"`; given a (debiased) coefficient vector for a selected subset of markers, this forms the linear index implied by those coefficients and estimates the residual treatment effect using kernel smoothing, as in the multiple marker, robust approach described in the `delta.s.estimate` documentation. Unlike `delta.s.estimate`, values outside the observed support are always extrapolated to the closest non-missing value.

Usage

```
.make.delta.s(coefs_vec, treat, control)
```

Arguments

<code>coefs_vec</code>	a named numeric vector of (debiased) coefficients for the selected markers; if of length zero (no markers selected), the residual treatment effect reduces to the overall treatment effect.
<code>treat</code>	data frame of treated observations, including a column <code>y</code> and one column for each marker named in <code>coefs_vec</code> .
<code>control</code>	data frame of control observations, including a column <code>y</code> and one column for each marker named in <code>coefs_vec</code> .

Value

a numeric value; the estimated residual treatment effect for the selected markers.

Author(s)

Layla Parast

References

Parast, L., Freestone, G., Tarr, J. (2026). Treatment-Aware Regularization via Multi-Objective Optimization for Surrogate Selection.

Parast, L., McDermott, M., Tian, L. (2016). Robust estimation of the proportion of treatment effect explained by surrogate marker information. *Statistics in Medicine*, 35(10):1637-1653.

.smoothed.regularized *Selects surrogate markers using an unweighted regularized approach*

Description

Helper function used by `select.surrogate` with `selection = "rr"`; this fits an unweighted (ordinary) cross-validated lasso to the treated group to select a subset of candidate markers, refits an unpenalized (debiased) model on the selected markers, and estimates the residual treatment effect and proportion of treatment effect explained for the selected subset.

Usage

```
.smoothed.regularized(data, surrogate_names, nlambdas = 20,
  lambda_rule = "lambda.1se", var = FALSE, boot.num = 200)
```

Arguments

<code>data</code>	data frame of all observations (treated and control), including columns <code>y</code> , <code>treat</code> , and one column per candidate marker.
<code>surrogate_names</code>	character vector of the candidate marker column names in <code>data</code> .
<code>nlambdas</code>	the number of penalty values considered along the lasso path, default is 20.
<code>lambda_rule</code>	the cross-validation rule used to select the penalty; choices are "lambda.1se" or "lambda.min", default is "lambda.1se".
<code>var</code>	TRUE or FALSE; indicates whether a bootstrap variance estimate is requested, default is FALSE.
<code>boot.num</code>	number of bootstrap resamples used when <code>var = TRUE</code> , default is 200.

Value

A list is returned:

<code>selection</code>	a named numeric vector of the penalized lasso coefficients of the selected markers.
<code>delta</code>	the estimated overall treatment effect.
<code>delta.s</code>	the estimated residual treatment effect for the selected markers.
<code>pte</code>	the estimated proportion of treatment effect explained by the selected markers.
<code>var</code>	a list, as returned by <code>.bootstrap_var</code> , of bootstrap standard errors and confidence intervals for <code>delta</code> , <code>delta.s</code> , and <code>pte</code> ; NULL if <code>var = FALSE</code> .

Author(s)

Layla Parast

References

Parast, L., Freestone, G., Tarr, J. (2026). Treatment-Aware Regularization via Multi-Objective Optimization for Surrogate Selection.

.tar.approach	<i>Selects surrogate markers using treatment-aware regularization</i>
---------------	---

Description

Helper function used by select.surrogate with selection = "TAR"; this fits a weighted lasso to the treated group, in which each candidate marker's penalty is weighted by the inverse of its estimated standardized treatment effect, and selects the penalty via cross-validation within the treated group to be the sparsest point along the weighted lasso path whose estimated proportion of treatment effect explained falls no more than epsilon below that of the full candidate set. The selected markers are then refit using an unpenalized (debiased) model, and the residual treatment effect and proportion of treatment effect explained are estimated for the selected subset.

Usage

```
.tar.approach(data, folds = 5, epsilon = 0.05, nlambdas = 20, surrogate_names,
c.constant = 1, small.n = 1e-6, var = FALSE, boot.num = 200)
```

Arguments

data	data frame of all observations (treated and control), including columns y, treat, and one column per candidate marker.
folds	the number of cross-validation folds, within the treated group, used to select the penalty, default is 5.
epsilon	the tolerance used to select the penalty: the selected subset is the sparsest point along the weighted lasso path whose estimated proportion of treatment effect explained falls no more than epsilon below that of the full candidate set, default is 0.05.
nlambdas	the number of penalty values considered along the weighted lasso path, default is 20.
surrogate_names	character vector of the candidate marker column names in data.
c.constant	a constant controlling the strength of the treatment-aware penalty weighting, default is 1.
small.n	a small positive constant added to the estimated treatment effect magnitude of each candidate marker before inverting it to form the penalty weights, to avoid dividing by a value at or near zero, default is 1e-6.
var	TRUE or FALSE; indicates whether a bootstrap variance estimate is requested, default is FALSE.
boot.num	number of bootstrap resamples used when var = TRUE, default is 200.

Value

A list is returned:

selection	a named numeric vector of the penalized (weighted) lasso coefficients of the selected markers.
delta	the estimated overall treatment effect.
delta.s	the estimated residual treatment effect for the selected markers.
pte	the estimated proportion of treatment effect explained by the selected markers.
var	a list, as returned by .bootstrap_var, of bootstrap standard errors and confidence intervals for delta, delta.s, and pte; NULL if var = FALSE.

Author(s)

Layla Parast

References

Parast, L., Freestone, G., Tarr, J. (2026). Treatment-Aware Regularization via Multi-Objective Optimization for Surrogate Selection.

.te.calculate	<i>Calculates standardized treatment effect magnitude for one candidate marker</i>
---------------	--

Description

Helper function used by select.surrogate with selection = "TAR"; this calculates the absolute standardized difference in means between the treated and control groups for one candidate surrogate marker, used to construct the treatment-aware penalty weights.

Usage

```
.te.calculate(s, treat, control)
```

Arguments

s	character; the name of the candidate marker column in treat and control.
treat	data frame of treated observations, containing a column named s.
control	data frame of control observations, containing a column named s.

Value

the absolute value of the difference in means between treat and control for marker s, divided by the pooled standard deviation.

Author(s)

Layla Parast

References

Parast, L., Freestone, G., Tarr, J. (2026). Treatment-Aware Regularization via Multi-Objective Optimization for Surrogate Selection.

Aug.R.s.surv.estimate *Calculates the augmented estimator of the proportion of treatment effect explained by the surrogate marker information measured at a specified time and primary outcome information up to that specified time*

Description

This function calculates the augmented version of the proportion of treatment effect on the primary outcome explained by the surrogate marker information measured at t_0 and primary outcome information up to t_0 . Variance estimates and 95 % confidence intervals for the augmented estimates are provided automatically; three versions of the confidence interval are provided: a normal approximation based interval, a quantile based interval and Fieller's confidence interval, all using perturbation-resampling. The user can also request an estimate of the incremental value of surrogate marker information.

Usage

```
Aug.R.s.surv.estimate(xone, xzero, deltaone, deltazero, sone, szero, t,
  weight.perturb = NULL, landmark, extrapolate = FALSE, transform = FALSE,
  basis.delta.one, basis.delta.zero, basis.delta.s.one = NULL,
  basis.delta.s.zero = NULL, incremental.value = FALSE, approx = T)
```

Arguments

xone	numeric vector, the observed event times in the treatment group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
xzero	numeric vector, the observed event times in the control group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
deltaone	numeric vector, the event indicators for the treatment group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
deltazero	numeric vector, the event indicators for the control group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
sone	numeric vector; surrogate marker measurement at t_0 for treated observations, assumed to be continuous. If $X_{1i} < t_0$, then the surrogate marker measurement should be NA.
szero	numeric vector; surrogate marker measurement at t_0 for control observations, assumed to be continuous. If $X_{1i} < t_0$, then the surrogate marker measurement should be NA.
t	the time of interest.
weight.perturb	weights used for perturbation resampling.
landmark	the landmark time t_0 or time of surrogate marker measurement.
extrapolate	TRUE or FALSE; indicates whether the user wants to use extrapolation.

transform	TRUE or FALSE; indicates whether the user wants to use a transformation for the surrogate marker.
basis.delta.one	either a vector of length n_1 or a matrix with n_1 rows; this is the basis transformation used for augmentation of $\hat{\Delta}(t)$ for treated observations only, all values must be numeric
basis.delta.zero	either a vector of length n_0 or a matrix with n_0 rows; this is the basis transformation used for augmentation of $\hat{\Delta}(t)$ for control observations only, all values must be numeric
basis.delta.s.one	either a vector of length n_1 or a matrix with n_1 rows; this is the basis transformation used for augmentation of $\hat{\Delta}_S(t, t_0)$ for treated observations only, all values must be numeric; default is to assume this is the same as basis.delta.one i.e. that the same basis transformation is used for both quantities
basis.delta.s.zero	either a vector of length n_0 or a matrix with n_0 rows; this is the basis transformation used for augmentation of $\hat{\Delta}_S(t, t_0)$ for control observations only, all values must be numeric; default is to assume this is the same as basis.delta.zero i.e. that the same basis transformation is used for both quantities
incremental.value	TRUE or FALSE; indicates whether the user would like to see the incremental value of the surrogate marker information, default is FALSE.
approx	TRUE or FALSE indicating whether an approximation should be used when calculating the probability of censoring; most relevant in settings where the survival time of interest for the primary outcome is greater than the last observed event but before the last censored case, default is TRUE.

Details

Please see R.s.surv.estimate documentation for details about the estimates before augmentation is performed. Recent work has shown that augmentation can lead to improvements in efficiency by taking advantage of the association between baseline information, denoted here as Z , and the primary outcome. This function calculates the augmented estimates of the quantities of interest. For example, the augmented version of $\hat{\Delta}(t)$ is defined as:

$$\hat{\Delta}(t)^{AUG} = \hat{\Delta}(t) + \gamma \left\{ n_1^{-1} \sum_{i=1}^{n_1} h(Z_{1i}) - n_0^{-1} \sum_{i=1}^{n_0} h(Z_{0i}) \right\}$$

where $Z_{gi}, i = 1, 2, \dots, n_g$ are i.i.d. random vectors of baseline covariates from treatment group g and $h(\cdot)$ is a basis transformation given a priori. Due to treatment randomization, $\{n_1^{-1} \sum_{i=1}^{n_1} h(Z_{1i}) - n_0^{-1} \sum_{i=1}^{n_0} h(Z_{0i})\}$ converges to zero in probability as the sample size goes to infinity and thus the augmented estimator converges to the same limit as the original counterparts. The quantity γ is selected such that the variance of $\hat{\Delta}(t)^{AUG}$ is minimized. That is, $\gamma = (\Xi_{12})(\Xi_{22})^{-1}$ where

$$\Xi_{12} = \text{cov}\left\{ \hat{\Delta}(t), n_1^{-1} \sum_{i=1}^{n_1} h(Z_{1i}) - n_0^{-1} \sum_{i=1}^{n_0} h(Z_{0i}) \right\},$$

$$\Xi_{22} = \text{var}\left\{ n_1^{-1} \sum_{i=1}^{n_1} h(Z_{1i}) - n_0^{-1} \sum_{i=1}^{n_0} h(Z_{0i}) \right\}$$

and thus we can obtain $\hat{\Delta}(t)^{AUG}$ by replacing γ with a consistent estimator, $\hat{\gamma}$ obtained using perturbation-resampling. A similar approach is used to obtain $\hat{\Delta}_S(t)^{AUG}$ and thus construct

$$\hat{R}_S(t, t_0)^{AUG} = 1 - \frac{\hat{\Delta}_S(t, t_0)^{AUG}}{\hat{\Delta}(t)^{AUG}}.$$

When only a single Z_{gi} is provided in the basis argument, the following basis is used in this function: $h(Z_{gi}) = (1, Z_{gi}, Z_{gi}^2)'$.

Value

A list is returned:

aug.delta	the estimate, $\hat{\Delta}(t)^{AUG}$.
aug.delta.s	the estimate, $\hat{\Delta}_S(t, t_0)^{AUG}$.
aug.R.s	the estimate, $\hat{R}_S(t, t_0)^{AUG}$.
aug.delta.var	the variance estimate of $\hat{\Delta}(t)^{AUG}$.
aug.delta.s.var	the variance estimate of $\hat{\Delta}_S(t, t_0)^{AUG}$.
aug.R.s.var	the variance estimate of $\hat{R}_S(t, t_0)^{AUG}$.
conf.int.normal.aug.delta	a vector of size 2; the 95% confidence interval for $\hat{\Delta}(t)^{AUG}$ based on a normal approximation.
conf.int.quantile.aug.delta	a vector of size 2; the 95% confidence interval for $\hat{\Delta}(t)^{AUG}$ based on sample quantiles of the perturbed values.
conf.int.normal.aug.delta.s	a vector of size 2; the 95% confidence interval for $\hat{\Delta}_S(t, t_0)^{AUG}$ based on a normal approximation.
conf.int.quantile.aug.delta.s	a vector of size 2; the 95% confidence interval for $\hat{\Delta}_S(t, t_0)^{AUG}$ based on sample quantiles of the perturbed values.
conf.int.normal.R.s	a vector of size 2; the 95% confidence interval for $\hat{R}_S(t, t_0)^{AUG}$ based on a normal approximation.
conf.int.quantile.aug.R.s	a vector of size 2; the 95% confidence interval for $\hat{R}_S(t, t_0)^{AUG}$ based on sample quantiles of the perturbed values..
conf.int.fieller.aug.R.s	a vector of size 2; the 95% confidence interval for $\hat{R}_S(t, t_0)^{AUG}$ based on Fieller's approach.
aug.delta.t	the estimate, $\hat{\Delta}_T(t, t_0)^{AUG}$; if incremental.vaue = TRUE.
aug.R.t	the estimate, $\hat{R}_T(t, t_0)^{AUG}$; if incremental.vaue = TRUE.
aug.incremental.value	the estimate, $\hat{IV}_S(t, t_0)^{AUG}$; if incremental.vaue = TRUE.
aug.delta.t.var	the variance estimate of $\hat{\Delta}_T(t, t_0)^{AUG}$; if incremental.vaue = TRUE.
aug.R.t.var	the variance estimate of $\hat{R}_T(t, t_0)^{AUG}$; if incremental.vaue = TRUE.

`aug.incremental.value.var`
the variance estimate of $\hat{IV}_S(t, t_0)^{AUG}$; if `incremental.vaue = TRUE`.

`aug.conf.int.normal.delta.t`
a vector of size 2; the 95% confidence interval for $\hat{\Delta}_T(t, t_0)^{AUG}$ based on a normal approximation; if `incremental.vaue = TRUE`.

`aug.conf.int.quantile.delta.t`
a vector of size 2; the 95% confidence interval for $\hat{\Delta}_T(t, t_0)^{AUG}$ based on sample quantiles of the perturbed values; if `incremental.vaue = TRUE`.

`aug.conf.int.normal.R.t`
a vector of size 2; the 95% confidence interval for $\hat{R}_T(t, t_0)^{AUG}$ based on a normal approximation; if `incremental.vaue = TRUE`.

`aug.conf.int.quantile.R.t`
a vector of size 2; the 95% confidence interval for $\hat{R}_T(t, t_0)^{AUG}$ based on sample quantiles of the perturbed values; if `incremental.vaue = TRUE`.

`aug.conf.int.fieller.R.t`
a vector of size 2; the 95% confidence interval for $\hat{R}_T(t, t_0)^{AUG}$ based on Fieller's approach, described above; if `incremental.vaue = TRUE`.

`aug.conf.int.normal.iv`
a vector of size 2; the 95% confidence interval for $\hat{IV}_S(t, t_0)^{AUG}$ based on a normal approximation; if `incremental.vaue = TRUE`.

`aug.conf.int.quantile.iv`
a vector of size 2; the 95% confidence interval for $\hat{IV}_S(t, t_0)^{AUG}$ based on sample quantiles of the perturbed values; if `incremental.vaue = TRUE`.

Note

If the treatment effect is not significant, the user will receive the following message: "Warning: it looks like the treatment effect is not significant; may be difficult to interpret the residual treatment effect in this setting". If the observed support of the surrogate marker for the control group is outside the observed support of the surrogate marker for the treatment group, the user will receive the following message: "Warning: observed supports do not appear equal, may need to consider a transformation or extrapolation".

Author(s)

Layla Parast

References

- Tian L, Cai T, Zhao L, Wei L. On the covariate-adjusted estimation for an overall treatment difference with data from a randomized comparative clinical trial. *Biostatistics* 2012; 13(2): 256-273.
- Garcia TP, Ma Y, Yin G. Efficiency improvement in a class of survival models through model-free covariate incorporation. *Lifetime Data Analysis* 2011; 17(4): 552-565.
- Zhang M, Tsiatis AA, Davidian M. Improving efficiency of inferences in randomized clinical trials using auxiliary covariates. *Biometrics* 2008; 64(3): 707-715.
- Parast, L., Cai, T., & Tian, L. (2017). Evaluating surrogate marker information using censored data. *Statistics in Medicine*, 36(11), 1767-1782.

Examples

```
#computationally intensive
#Aug.R.s.surv.estimate(xone = d_example_surv$x1, xzero = d_example_surv$x0,
#deltaone = d_example_surv$delta1, deltazero = d_example_surv$delta0,
#sone = d_example_surv$s1, szero = d_example_surv$s0, t=3, landmark = 1,
#basis.delta.one = d_example_surv$z1 , basis.delta.zero = d_example_surv$z0)
```

augment.est.vector	<i>Augmentation function</i>
--------------------	------------------------------

Description

Augmentation function

Usage

```
augment.est.vector(point.delta, perturb.delta, treat.ind, basis, weights)
```

Arguments

point.delta
 perturb.delta
 treat.ind
 basis
 weights

Author(s)

Layla Parast

calculate.var.np	<i>calculates closed form variance estimate for R</i>
------------------	---

Description

calculates closed form variance estimate for R; used in R function that corrects for measurement error

Usage

```
calculate.var.np(s1, s0, y1, y0, extrapolate = TRUE)
```

Arguments

s1	numeric vector or matrix; surrogate marker for treated observations, assumed to be continuous. If there are multiple surrogates then this should be a matrix with n_1 (number of treated observations) rows and n.s (number of surrogate markers) columns.
s0	numeric vector; surrogate marker for control observations, assumed to be continuous. If there are multiple surrogates then this should be a matrix with n_0 (number of control observations) rows and n.s (number of surrogate markers) columns.
y1	numeric vector; primary outcome for treated observations, assumed to be continuous.
y0	numeric vector; primary outcome for control observations, assumed to be continuous.
extrapolate	TRUE or FALSE; indicates whether the user wants to use extrapolation.

Value

total	matrix needed for variance calculation
psionly	matrix needed for variance calculation

Author(s)

Layla Parast

References

- Parast, L., McDermott, M., Tian, L. (2016). Robust estimation of the proportion of treatment effect explained by surrogate marker information. *Statistics in Medicine*, 35(10):1637-1653.
- Parast, L., Garcia, T. P., Prentice, R. L., & Carroll, R. J. (2022). Robust methods to correct for measurement error when evaluating a surrogate marker. *Biometrics*, 78(1), 9-23.

censor.weight	<i>Calculates censoring probability for weighting</i>
---------------	---

Description

Helper function; calculates censoring probability needed for inverse probability of censoring weighting

Usage

```
censor.weight(data.x, data.delta, t, weight = NULL, approx = T)
```

Arguments

<code>data.x</code>	numeric vector, the observed event time: $X = \min(T, C)$ where T is the time of the primary outcome, C is the censoring time
<code>data.delta</code>	numeric vector of 0/1, the censoring indicator: $D = I(T < C)$ where T is the time of the primary outcome, C is the censoring time
<code>t</code>	number, the time of interest
<code>weight</code>	a numeric vector or matrix of weights used for perturbation-resampling, default is null.
<code>approx</code>	TRUE or FALSE indicating whether an approximation should be used when calculating the probability of censoring; most relevant in settings where the survival time of interest for the primary outcome is greater than the last observed event but before the last censored case, default is TRUE.

Details

Computes the Kaplan Meier estimate of survival for the censoring random variable at the specified time

Value

Kaplan Meier estimate of survival for censoring at time t

Author(s)

Layla Parast

cumsum2

Helper function

Description

Helper function; should not be called directly by user.

Usage

```
cumsum2(mydat)
```

Arguments

<code>mydat</code>	<code>mydat</code>
--------------------	--------------------

Value

<code>out</code>	matrix
------------------	--------

Author(s)

Layla Parast

delper.si	<i>Helper function</i>
-----------	------------------------

Description

Helper function; should not be called directly by user.

Usage

```
helper.si(yy,FUN,Yi,Vi=NULL)
```

Arguments

yy	yy
FUN	FUN
Yi	Yi
Vi	Vi

Value

out	matrix
-----	--------

Author(s)

Layla Parast

delta.estimate	<i>Calculates treatment effect</i>
----------------	------------------------------------

Description

This function calculates the treatment effect estimate, the difference in the average outcome in the treatment group minus the control group. This function is intended to be used for a fully observed continuous outcome. The user can also request a variance estimate, estimated using perturbing-resampling, and a 95% confidence interval. If a confidence interval is requested two versions are provided: a normal approximation based interval and a quantile based interval, both use perturbation-resampling.

Usage

```
delta.estimate(yone,yzero, var = FALSE, conf.int = FALSE, weight = NULL,
weight.perturb = NULL)
```

Arguments

yone	numeric vector; primary outcome for treated observations.
yzero	numeric vector; primary outcome for control observations.
var	TRUE or FALSE; indicates whether a variance estimate for delta is requested, default is FALSE.
conf.int	TRUE or FALSE; indicates whether a 95% confidence interval for delta is requested, default is FALSE.
weight	a n1+n0 by x matrix of weights where n1 = length of yone and n0 = length of yzero, default is null; generally not supplied by use but only used by other functions.
weight.perturb	a n1+n0 by x matrix of weights where n1 = length of yone and n0 = length of yzero, default is null; generally used for confidence interval construction and may be supplied by user.

Details

Let $Y^{(1)}$ and $Y^{(0)}$ denote the primary outcome under the treatment and primary outcome under the control, respectively. The treatment effect, Δ , is the expected difference in $Y^{(1)}$ compared to $Y^{(0)}$, $\Delta = E(Y^{(1)} - Y^{(0)})$. We estimate Δ as

$$\hat{\Delta} = n_1^{-1} \sum_{i=1}^{n_1} Y_{1i} - n_0^{-1} \sum_{i=1}^{n_0} Y_{0i}$$

where Y_{1i} is the observed primary outcome for person i in the treated group, Y_{0i} is the observed primary outcome for person i in the control group, and n_1 and n_0 are the number of individuals in the treatment and control group, respectively. Randomized treatment assignment is assumed throughout this package.

Variance estimation and confidence interval construction are performed using perturbation-resampling. Specifically, let $\{V^{(b)} = (V_{11}^{(b)}, \dots, V_{1n_1}^{(b)}, V_{01}^{(b)}, \dots, V_{0n_0}^{(b)})^T, b = 1, \dots, D\}$ be $n \times D$ independent copies of a positive random variables V from a known distribution with unit mean and unit variance. Let

$$\hat{\Delta}^{(b)} = \frac{\sum_{i=1}^{n_1} V_{1i}^{(b)} Y_{1i}}{\sum_{i=1}^{n_1} V_{1i}^{(b)}} - \frac{\sum_{i=1}^{n_0} V_{0i}^{(b)} Y_{0i}}{\sum_{i=1}^{n_0} V_{0i}^{(b)}}.$$

The variance of $\hat{\Delta}$ is obtained as the empirical variance of $\{\hat{\Delta}^{(b)}, b = 1, \dots, D\}$. In this package, we use weights generated from an Exponential(1) distribution and use $D = 500$. We construct two versions of the 95% confidence interval for $\hat{\Delta}$: one based on a normal approximation confidence interval using the estimated variance and another taking the 2.5th and 97.5th empirical percentiles of $\hat{\Delta}^{(b)}$.

Value

A list is returned:

delta	the estimate, $\hat{\Delta}$, described above.
var	the variance estimate of $\hat{\Delta}$; if var = TRUE or conf.int = TRUE.
conf.int.normal	a vector of size 2; the 95% confidence interval for $\hat{\Delta}$ based on a normal approximation; if conf.int = TRUE.
conf.int.quantile	a vector of size 2; the 95% confidence interval for $\hat{\Delta}$ based on sample quantiles of the perturbed values, described above; if conf.int = TRUE.

Author(s)

Layla Parast

Examples

```
data(d_example)
names(d_example)
delta.estimate(yone=d_example$y1, yzero=d_example$y0)
```

delta.multiple.surv	<i>Calculates robust residual treatment effect accounting for multiple surrogate markers at a specified time and primary outcome information up to that specified time</i>
---------------------	--

Description

This function calculates the robust estimate of the residual treatment effect accounting for multiple surrogate markers measured at t_0 and primary outcome information up to t_0 i.e. the hypothetical treatment effect if both the surrogate marker distributions at t_0 and survival up to t_0 in the treatment group look like the surrogate marker distributions and survival up to t_0 in the control group. Ideally this function is only used as a helper function and is not directly called.

Usage

```
delta.multiple.surv(xone, xzero, deltaone, deltazero, sone, szero, type = 1, t,
weight.perturb = NULL, landmark, extrapolate = FALSE, transform = FALSE,
approx = T)
```

Arguments

xone	numeric vector, the observed event times in the treatment group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
xzero	numeric vector, the observed event times in the control group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
deltaone	numeric vector, the event indicators for the treatment group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
deltazero	numeric vector, the event indicators for the control group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
sone	matrix of numeric values; surrogate marker measurements at t_0 for treated observations. If $X_{1i} < t_0$, then the surrogate marker measurements should be NA.
szero	matrix of numeric values; surrogate marker measurements at t_0 for control observations. If $X_{0i} < t_0$, then the surrogate marker measurements should be NA.
type	type of estimate; options are 1 = two-stage robust estimator, 2 = weighted two-stage robust estimator, 3 = double-robust estimator, 4 = two-stage model-based estimator, 5 = weighted estimator, 6 = double-robust model-based estimator; default is 1.
t	the time of interest.

weight.perturb	weights used for perturbation resampling.
landmark	the landmark time t_0 or time of surrogate marker measurement.
extrapolate	TRUE or FALSE; indicates whether the user wants to use extrapolation.
transform	TRUE or FALSE; indicates whether the user wants to use a transformation for the surrogate marker psuedo-score.
approx	TRUE or FALSE indicating whether an approximation should be used when calculating the probability of censoring; most relevant in settings where the survival time of interest for the primary outcome is greater than the last observed event but before the last censored case, default is TRUE.

Details

Details are included in the documentation for R.multiple.surv.

Value

$\hat{\Delta}_S(t, t_0)$, the residual treatment effect estimate accounting for multiple surrogarte markers measured at t_0 and primary outcome information up to t_0 .

Note

If the treatment effect is not significant, the user will receive the following message: "Warning: it looks like the treatment effect is not significant; may be difficult to interpret the residual treatment effect in this setting".

Author(s)

Layla Parast

References

Parast, L., Cai, T., & Tian, L. (2021). Evaluating multiple surrogate markers with censored data. *Biometrics*, 77(4), 1315-1327.

Examples

```
data(d_example_multiple)
names(d_example_multiple)
## Not run:
delta.multiple.surv(xone = d_example_multiple$x1, xzero = d_example_multiple$x0, deltaone =
  d_example_multiple$delta1, deltazero = d_example_multiple$delta0, sone =
  as.matrix(d_example_multiple$s1), szero = as.matrix(d_example_multiple$s0),
  type =1, t = 1, landmark=0.5)

## End(Not run)
```

delta.s.estimate	<i>Calculates model-based or robust residual treatment effect</i>
------------------	---

Description

This function calculates the model-based or robust estimate of the residual treatment effect i.e. the hypothetical treatment effect if the distribution of the surrogate in the treatment group looks like the distribution of the surrogate in the control group. Ideally, this function is only used as a helper function and is not directly called.

Usage

```
delta.s.estimate(sone, szero, yone, yzero, weight.perturb = NULL, number="single",
type="robust", warn.te = FALSE, warn.support = FALSE, extrapolate = FALSE,
transform = FALSE)
```

Arguments

sone	numeric vector or matrix; surrogate marker for treated observations, assumed to be continuous. If there are multiple surrogates then this should be a matrix with n_1 (number of treated observations) rows and n.s (number of surrogate markers) columns.
szero	numeric vector or matrix; surrogate marker for control observations, assumed to be continuous. If there are multiple surrogates then this should be a matrix with n_0 (number of control observations) rows and n.s (number of surrogate markers) columns.
yone	numeric vector; primary outcome for treated observations.
yzero	numeric vector; primary outcome for control observations.
weight.perturb	a $n_1 + n_0$ by x matrix of weights where n_1 = length of yone and n_0 = length of yzero; generally used for variance estimation and confidence interval construction, default is null.
number	specifies the number of surrogate markers; choices are "multiple" or "single", default is "single".
type	specifies the type of estimation; choices are "robust" or "model", default is "robust".
warn.te	value passed from R.s.estimate function to control warnings; user does not need to specify.
warn.support	value passed from R.s.estimate function to control warnings; user does not need to specify.
extrapolate	TRUE or FALSE; indicates whether the user wants to use extrapolation.
transform	TRUE or FALSE; indicates whether the user wants to use a transformation for the surrogate marker.

Details

Details are included in the documentation for R.s.estimate.

Value

$\hat{\Delta}_S$, the model-based or robust residual treatment effect estimate.

Note

If the treatment effect is not significant, the user will receive the following message: "Warning: it looks like the treatment effect is not significant; may be difficult to interpret the residual treatment effect in this setting". In the single marker case with the robust estimation approach, if the observed support of the surrogate marker for the control group is outside the observed support of the surrogate marker for the treatment group, the user will receive the following message: "Warning: observed supports do not appear equal, may need to consider a transformation or extrapolation".

Author(s)

Layla Parast

References

Parast, L., McDermott, M., Tian, L. (2015). Robust estimation of the proportion of treatment effect explained by surrogate marker information. *Statistics in Medicine*, 35(10):1637-1653.

Wang, Y., & Taylor, J. M. (2002). A measure of the proportion of treatment effect explained by a surrogate marker. *Biometrics*, 58(4), 803-812.

Examples

```
data(d_example)
names(d_example)
delta.s.estimate(yone=d_example$y1, yzero=d_example$y0, sone=d_example$s1.a, szero=
d_example$s0.a, number = "single", type = "robust")
delta.s.estimate(yone=d_example$y1, yzero=d_example$y0, sone=d_example$s1.a, szero=
d_example$s0.a, number = "single", type = "model")
delta.s.estimate(yone=d_example$y1, yzero=d_example$y0, sone=cbind(d_example$s1.a,
d_example$s1.b, d_example$s1.c), szero=cbind(d_example$s0.a, d_example$s0.b, d_example$s0.c),
number = "multiple", type = "robust")
delta.s.estimate(yone=d_example$y1, yzero=d_example$y0, sone=cbind(d_example$s1.a,
d_example$s1.b, d_example$s1.c), szero=cbind(d_example$s0.a, d_example$s0.b, d_example$s0.c),
number = "multiple", type = "model")
```

delta.s.surv.estimate	<i>Calculates robust residual treatment effect accounting for surrogate marker information measured at a specified time and primary outcome information up to that specified time</i>
-----------------------	---

Description

This function calculates the robust estimate of the residual treatment effect accounting for surrogate marker information measured at t_0 and primary outcome information up to t_0 i.e. the hypothetical treatment effect if both the surrogate marker distribution at t_0 and survival up to t_0 in the treatment group look like the surrogate marker distribution and survival up to t_0 in the control group. Ideally this function is only used as a helper function and is not directly called.

Usage

```
delta.s.surv.estimate(xone, xzero, deltaone, deltazero, sone, szero, t,
  weight.perturb = NULL, landmark, extrapolate = FALSE, transform = FALSE,
  approx = T, warn.te = FALSE, warn.support = FALSE)
```

Arguments

xone	numeric vector, the observed event times in the treatment group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
xzero	numeric vector, the observed event times in the control group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
deltaone	numeric vector, the event indicators for the treatment group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
deltazero	numeric vector, the event indicators for the control group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
sone	numeric vector; surrogate marker measurement at t_0 for treated observations, assumed to be continuous. If $X_{1i} < t_0$, then the surrogate marker measurement should be NA.
szero	numeric vector; surrogate marker measurement at t_0 for control observations, assumed to be continuous. If $X_{1i} < t_0$, then the surrogate marker measurement should be NA.
t	the time of interest.
weight.perturb	weights used for perturbation resampling.
landmark	the landmark time t_0 or time of surrogate marker measurement.
extrapolate	TRUE or FALSE; indicates whether the user wants to use extrapolation.
transform	TRUE or FALSE; indicates whether the user wants to use a transformation for the surrogate marker.
approx	TRUE or FALSE indicating whether an approximation should be used when calculating the probability of censoring; most relevant in settings where the survival time of interest for the primary outcome is greater than the last observed event but before the last censored case, default is TRUE.
warn.te	value passed from R.s.estimate function to control warnings; user does not need to specify.
warn.support	value passed from R.s.estimate function to control warnings; user does not need to specify.

Details

Details are included in the documentation for R.s.surv.estimate.

Value

$\hat{\Delta}_S(t, t_0)$, the robust residual treatment effect estimate accounting for surrogate marker information measured at t_0 and primary outcome information up to t_0 .

Note

If the treatment effect is not significant, the user will receive the following message: "Warning: it looks like the treatment effect is not significant; may be difficult to interpret the residual treatment effect in this setting". If the observed support of the surrogate marker for the control group is outside the observed support of the surrogate marker for the treatment group, the user will receive the following message: "Warning: observed supports do not appear equal, may need to consider a transformation or extrapolation".

Author(s)

Layla Parast

References

Parast, L., Cai, T., & Tian, L. (2017). Evaluating surrogate marker information using censored data. *Statistics in Medicine*, 36(11), 1767-1782.

Examples

```
data(d_example_surv)
names(d_example_surv)
```

delta.surv.estimate	<i>Calculates treatment effect in a survival setting</i>
---------------------	--

Description

This function calculates the treatment effect in the survival setting i.e. the difference in survival at time t between the treatment group and the control group. The user can also request a variance estimate, estimated using perturbing-resampling, and a 95% confidence interval. If a confidence interval is requested two versions are provided: a normal approximation based interval and a quantile based interval, both use perturbation-resampling.

Usage

```
delta.surv.estimate(xone, xzero, deltaone, deltazero, t, var = FALSE, conf.int = FALSE, weight = NULL, weight.perturb = NULL, approx = T)
```

Arguments

xone	numeric vector, the observed event times in the treatment group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
xzero	numeric vector, the observed event times in the control group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
deltaone	numeric vector, the event indicators for the treatment group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
deltazero	numeric vector, the event indicators for the control group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
t	the time of interest.

var	TRUE or FALSE; indicates whether a variance estimate for delta is requested, default is FALSE.
conf.int	TRUE or FALSE; indicates whether a 95% confidence interval for delta is requested, default is FALSE.
weight	a $n_1 + n_0$ by x matrix of weights where n_1 = sample size in treatment group and n_0 = sample size in the control group, default is null; generally not supplied by use but only used by other functions.
weight.perturb	a $n_1 + n_0$ by x matrix of weights where n_1 = sample size in treatment group and n_0 = sample size in the control group, default is null; generally used for confidence interval construction and may be supplied by user.
approx	TRUE or FALSE indicating whether an approximation should be used when calculating the probability of censoring; most relevant in settings where the survival time of interest for the primary outcome is greater than the last observed event but before the last censored case, default is TRUE.

Details

Let G be the binary treatment indicator with $G = 1$ for treatment and $G = 0$ for control and we assume throughout that subjects are randomly assigned to a treatment group at baseline. Let T denote the time of the primary outcome of interest, death for example. We use potential outcomes notation such that $T^{(g)}$ denotes the time of the primary outcome under treatment $G = g$. We define the treatment effect, $\Delta(t)$, as the difference in survival rates by time t under treatment versus control,

$$\Delta(t) = E\{I(T^{(1)} > t)\} - E\{I(T^{(0)} > t)\} = P(T^{(1)} > t) - P(T^{(0)} > t)$$

where $t > t_0$

Due to censoring, our data consist of n_1 observations $\{(X_{1i}, \delta_{1i}), i = 1, \dots, n_1\}$ from the treatment group $G = 1$ and n_0 observations $\{(X_{0i}, \delta_{0i}), i = 1, \dots, n_0\}$ from the control group $G = 0$ where $X_{gi} = \min(T_{gi}, C_{gi})$, $\delta_{gi} = I(T_{gi} < C_{gi})$, and C_{gi} denotes the censoring time for $g = 1, 0$, for individual i . Throughout, we estimate the treatment effect $\Delta(t)$ as

$$\hat{\Delta}(t) = n_1^{-1} \sum_{i=1}^{n_1} \frac{I(X_{1i} > t)}{\hat{W}_1^C(t)} - n_0^{-1} \sum_{i=1}^{n_0} \frac{I(X_{0i} > t)}{\hat{W}_0^C(t)}$$

where $\hat{W}_g^C(\cdot)$ is the Kaplan-Meier estimator of survival for censoring for $g = 1, 0$.

Variance estimation and confidence interval construction are performed using perturbation-resampling. Specifically, let $\{V^{(b)} = (V_{11}^{(b)}, \dots, V_{1n_1}^{(b)}, V_{01}^{(b)}, \dots, V_{0n_0}^{(b)})^T, b = 1, \dots, D\}$ be $n \times D$ independent copies of a positive random variables V from a known distribution with unit mean and unit variance. Let

$$\hat{\Delta}^{(b)}(t) = \frac{\sum_{i=1}^{n_1} V_{1i}^{(b)} I(X_{1i} > t)}{\sum_{i=1}^{n_1} V_{1i}^{(b)} \hat{W}_1^{C(b)}(t)} - \frac{\sum_{i=1}^{n_0} V_{0i}^{(b)} I(X_{0i} > t)}{\sum_{i=1}^{n_0} V_{0i}^{(b)} \hat{W}_0^{C(b)}(t)}.$$

In this package, we use weights generated from an Exponential(1) distribution and use $D = 500$. The variance of $\hat{\Delta}(t)$ is obtained as the empirical variance of $\{\hat{\Delta}(t)^{(b)}, b = 1, \dots, D\}$. We construct two versions of the 95% confidence interval for $\hat{\Delta}(t)$: one based on a normal approximation confidence interval using the estimated variance and another taking the 2.5th and 97.5th empirical percentiles of $\hat{\Delta}(t)^{(b)}$.

Value

A list is returned:

delta	the estimate, $\hat{\Delta}(t)$, described above.
var	the variance estimate of $\hat{\Delta}(t)$; if var = TRUE or conf.int = TRUE.
conf.int.normal	a vector of size 2; the 95% confidence interval for $\hat{\Delta}(t)$ based on a normal approximation; if conf.int = TRUE.
conf.int.quantile	a vector of size 2; the 95% confidence interval for $\hat{\Delta}(t)$ based on sample quantiles of the perturbed values, described above; if conf.int = TRUE.

Author(s)

Layla Parast

Examples

```
data(d_example_surv)
names(d_example_surv)
delta.surv.estimate(xone = d_example_surv$x1, xzero = d_example_surv$x0,
deltaone = d_example_surv$delta1, deltazero = d_example_surv$delta0, t = 3)
```

delta.t.surv.estimate *Calculates robust residual treatment effect accounting only for primary outcome information up to a specified time*

Description

This function calculates the robust estimate of the residual treatment effect accounting only for primary outcome information up to t_0 i.e. the hypothetical treatment effect if survival up to t_0 in the treatment group looks like survival up to t_0 in the control group. Ideally this function is only used as a helper function and is not directly called.

Usage

```
delta.t.surv.estimate(xone, xzero, deltaone, deltazero, t, weight.perturb = NULL,
landmark, approx = T)
```

Arguments

xone	numeric vector, the observed event times in the treatment group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
xzero	numeric vector, the observed event times in the control group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
deltaone	numeric vector, the event indicators for the treatment group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
deltazero	numeric vector, the event indicators for the control group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
t	the time of interest.

weight.perturb	weights used for perturbation resampling.
landmark	the landmark time t_0 or time of surrogate marker measurement.
approx	TRUE or FALSE indicating whether an approximation should be used when calculating the probability of censoring; most relevant in settings where the survival time of interest for the primary outcome is greater than the last observed event but before the last censored case, default is TRUE.

Details

Details are included in the documentation for `R.t.surv.estimate`.

Value

$\hat{\Delta}_T(t, t_0)$, the robust residual treatment effect estimate accounting only for survival up to t_0 .

Note

If the treatment effect is not significant, the user will receive the following message: "Warning: it looks like the treatment effect is not significant; may be difficult to interpret the residual treatment effect in this setting".

Author(s)

Layla Parast

References

Parast, L., Cai, T., & Tian, L. (2017). Evaluating surrogate marker information using censored data. *Statistics in Medicine*, 36(11), 1767-1782.

Examples

```
data(d_example_surv)
names(d_example_surv)
```

d_example

Hypothetical data

Description

Hypothetical data to be used in examples.

Usage

```
data(d_example)
```

Format

A list with 8 elements representing 500 observations from a control group and 500 observations from a treatment group:

- s1.a First surrogate marker measurement for treated observations.
- s1.b Second surrogate marker measurement for treated observations.
- s1.c Third surrogate marker measurement for treated observations.
- y1 Primary outcome for treated observations.
- s0.a First surrogate marker measurement for control observations.
- s0.b Second surrogate marker measurement for control observations.
- s0.c Third surrogate marker measurement for control observations.
- y0 Primary outcome for control observations.

Examples

```
data(d_example)
names(d_example)
```

d_example_me

Hypothetical data with replicate measurements

Description

Hypothetical data to be used in measurement error example.

Usage

```
data(d_example_me)
```

Format

A list with 10 elements representing 500 observations from a control group and 500 observations from a treatment group:

- y1 Primary outcome for treated observations.
- s1 Surrogate marker for treated observations.
- s1_rep1 Replicate measurement of the surrogate marker for treated observations.
- s1_rep2 Replicate measurement of the surrogate marker for treated observations.
- s1_rep3 Replicate measurement of the surrogate marker for treated observations.
- y0 Primary outcome for control observations.
- s0 Surrogate marker for control observations.
- s0_rep1 Replicate measurement of the surrogate marker for control observations.
- s0_rep2 Replicate measurement of the surrogate marker for control observations.
- s0_rep3 Replicate measurement of the surrogate marker for control observations.

Examples

```
data(d_example_me)
names(d_example_me)
```

d_example_multiple	<i>Hypothetical survival data with multiple surrogate markers</i>
--------------------	---

Description

Hypothetical survival data with multiple surrogate markers to be used in examples.

Usage

```
data(d_example_multiple)
```

Format

A list with 6 elements representing 1000 observations from a control group and 1000 observations from a treatment group:

- s1 Surrogate marker measurements for treated observations; these markers are measured at time = 0.5. For observations that experience the primary outcome or are censored before 0.5, the surrogate values are NA.
- x1 The observed event or censoring time for treated observations; $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
- delta1 The indicator identifying whether the treated observation was observed to have the event or was censored; $D = 1 * (T < C)$ where T is the time of the primary outcome and C is the censoring time.
- s0 Surrogate marker measurements for control observations; these markers are measured at time = 0.5. For observations that experience the primary outcome or are censored before 0.5, the surrogate values are NA.
- x0 The observed event or censoring time for control observations; $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
- delta0 The indicator identifying whether the control observation was observed to have the event or was censored; $D = 1 * (T < C)$ where T is the time of the primary outcome and C is the censoring time.

Examples

```
data(d_example_multiple)
names(d_example_multiple)
```

d_example_surv	<i>Hypothetical survival data</i>
----------------	-----------------------------------

Description

Hypothetical survival data to be used in examples.

Usage

```
data(d_example_surv)
```

Format

A list with 8 elements representing 500 observations from a control group and 500 observations from a treatment group:

s1 Surrogate marker measurement for treated observations; this marker is measured at time = 0.5. For observations that experience the primary outcome or are censored before 0.5, this value is NA.

x1 The observed event or censoring time for treated observations; $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.

delta1 The indicator identifying whether the treated observation was observed to have the event or was censored; $D = 1 * (T < C)$ where T is the time of the primary outcome and C is the censoring time.

s0 Surrogate marker measurement for control observations; this marker is measured at time = 0.5. For observations that experience the primary outcome or are censored before 0.5, this value is NA.

x0 The observed event or censoring time for control observations; $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.

delta0 The indicator identifying whether the control observation was observed to have the event or was censored; $D = 1 * (T < C)$ where T is the time of the primary outcome and C is the censoring time.

z1 A baseline covariate value for treated observations.

z0 A baseline covariate value for control observations.

Examples

```
data(d_example_surv)
names(d_example_surv)
```

d_tar_example

Example data for treatment-aware regularization

Description

One simulated dataset, generated from Setting 1 of the simulation study of Parast, Freestone, and Tarr (2026), to be used in the examples for select.surrogate. Nine correlated candidate surrogate markers are generated for each observation; the first three (s1, s2, s3) both shift between treatment groups and drive the outcome, while the remaining six are not associated with the treatment effect on the outcome.

Usage

```
data(d_tar_example)
```

Format

A list with 4 elements representing 500 observations from a treatment group and 500 observations from a control group, each with 9 candidate surrogate markers:

sone a 500 by 9 numeric matrix of candidate surrogate markers for the treated observations, with columns named s1, ..., s9.

szero a 500 by 9 numeric matrix of candidate surrogate markers for the control observations, with columns named s1, ..., s9.

yone a numeric vector of length 500; the primary outcome for the treated observations.

yzero a numeric vector of length 500; the primary outcome for the control observations.

References

Parast, L., Freestone, G., Tarr, J. (2026). Treatment-Aware Regularization via Multi-Objective Optimization for Surrogate Selection.

Examples

```
data(d_tar_example)
names(d_tar_example)
```

fieller.calculate.me *Constructs Fieller's confidence interval.*

Description

Constructs Fieller's confidence interval - closed form, not using perturbation.

Usage

```
fieller.calculate.me(a, B, var.mat)
```

Arguments

a	value of numerator
B	value of denominator
var.mat	variance-covariance matrix for a and B

Details

Produces a confidence interval for $1-a/B$ when closed form variance and covariance estimates are available. See documentation for R.s.estimate.me for more detail.

Value

Returns a vector of length 2, lower bound of the 95% confidence interval and upper bound of the 95% confidence interval.

Author(s)

Layla Parast

References

- Parast, L., Garcia, T. P., Prentice, R. L., & Carroll, R. J. (2022). Robust methods to correct for measurement error when evaluating a surrogate marker. *Biometrics*, 78(1), 9-23.
- Fieller, Edgar C. (1954). Some problems in interval estimation. *Journal of the Royal Statistical Society. Series B (Methodological)*, 175-185.
- Fieller, E. C. (1940). The biological standardization of insulin. *Supplement to the Journal of the Royal Statistical Society*, 1-64.

fieller.ci	<i>Constructs Fieller's confidence interval.</i>
------------	--

Description

Constructs Fieller's confidence interval.

Usage

```
fieller.ci(perturb.delta.s, perturb.delta, delta.s, delta)
```

Arguments

perturb.delta.s	numeric vector; the perturbed values for $\hat{\Delta}_S$, the residual treatment effect estimate, used in variance estimation and confidence interval construction.
perturb.delta	numeric vector; the perturbed values for $\hat{\Delta}$, the treatment effect estimate, used in variance estimation and confidence interval construction.
delta.s	the residual treatment effect, Δ_S , estimate, $\hat{\Delta}_S$.
delta	the treatment effect, Δ , estimate, $\hat{\Delta}$.

Details

See documentation for R.s.estimate for more detail.

Value

Returns a vector of length 2, lower bound of the 95% confidence interval and upper bound of the 95% confidence interval.

Author(s)

Layla Parast

References

- Fieller, Edgar C. (1954). Some problems in interval estimation. *Journal of the Royal Statistical Society. Series B (Methodological)*, 175-185.
- Fieller, E. C. (1940). The biological standardization of insulin. *Supplement to the Journal of the Royal Statistical Society*, 1-64.
- Parast, L., McDermott, M., Tian, L. (2016). Robust estimation of the proportion of treatment effect explained by surrogate marker information. *Statistics in Medicine*, 35(10):1637-1653.

Kern.FUN	<i>Calculates kernel matrix</i>
----------	---------------------------------

Description

Helper function; this calculates the kernel matrix

Usage

```
Kern.FUN(zz, zi, bw)
```

Arguments

zz	zz
zi	zi
bw	bandwidth

Value

the kernel matrix

Author(s)

Layla Parast

me.variance.estimate	<i>Estimates measurement error variance given replicate data.</i>
----------------------	---

Description

Estimates measurement error variance given replicate data using a simple components of variance analysis.

Usage

```
me.variance.estimate(replicates)
```

Arguments

replicates	matrix of data where each row indicates a subject and each column is a replicated measurement; columns can have NAs when subjects have different numbers of measurements.
------------	---

Details

Estimates measurement error variance given replicate data using a simple components of variance analysis.

Value

estimate of measurement error variance

Author(s)

Layla Parast

References

Carroll, R. J., Ruppert, D., Crainiceanu, C. M., and Stefanski, L. A. (2006). Measurement error in nonlinear models: a modern perspective. Chapman and Hall/CRC.

Parast, L., Garcia, T. P., Prentice, R. L., & Carroll, R. J. (2022). Robust methods to correct for measurement error when evaluating a surrogate marker. Biometrics, 78(1), 9-23.

perturb.nu.vector	<i>Helper for augmentation function</i>
-------------------	---

Description

Helper for augmentation function

Usage

```
perturb.nu.vector(mat, weights)
```

Arguments

mat

weights

pred.smooth	<i>Calculates expected outcome for control group surrogate values</i>
-------------	---

Description

Helper function; calculates the expected outcome for control group surrogate values using treatment group surrogate and outcome information.

Usage

```
pred.smooth(zz, zi.one, bw = NULL, y1, weight = NULL)
```

Arguments

zz	surrogate values in the treatment group; used to estimate conditional mean function
----	---

zi.one	surrogate values in the control group
--------	---------------------------------------

bw	bandwidth
----	-----------

y1	outcome in the treatment group
----	--------------------------------

weight	weight
--------	--------

Details

Details are included in the documentation for R.s.estimate

Value

expected outcome for each surrogate value in the control group

Author(s)

Layla Parast

References

Parast, L., McDermott, M., Tian, L. (2016). Robust estimation of the proportion of treatment effect explained by surrogate marker information. *Statistics in medicine*, 35(10):1637-1653.

pred.smooth.surv	<i>Calculates the conditional probability of survival for control group values</i>
------------------	--

Description

Helper function; calculates the estimated probability of survival for control group for control group surrogate values using treatment group surrogate and outcome information.

Usage

```
pred.smooth.surv(xone.f, deltaone.f, sone.f, szero.one, myt, weight.pred,
  extrapolate, transform, ps.weight = NULL, warn.support = FALSE)
```

Arguments

xone.f	observed event times in the treatment group
deltaone.f	censoring indicators in the treatment group
sone.f	surrogate marker values in the treatment group
szero.one	surrogate marker values in the control group
myt	time of interest
weight.pred	weight used for perturbation resampling
extrapolate	TRUE or FALSE; indicates whether local constant extrapolation should be used, default is FALSE
transform	TRUE or FALSE; indicates whether a transformation should be used, default is FALSE
ps.weight	vector of propensity score weights to be used in the multiple marker setting, default is NULL
warn.support	value passed from R.s.estimate function to control warnings; user does not need to specify.

Details

Details are included in the documentation for R.s.surv.estimate and R.multiple.surv

Value

conditional probability of survival past t for control group

Author(s)

Layla Parast

R.multiple.surv	<i>Calculates the proportion of treatment effect explained by multiple surrogate markers measured at a specified time and primary outcome information up to that specified time</i>
-----------------	---

Description

This function calculates the proportion of treatment effect on the primary outcome explained by multiple surrogate markers measured at t_0 and primary outcome information up to t_0 . The user can also request a variance estimate, estimated using perturbing-resampling, and a 95% confidence interval. If a confidence interval is requested three versions are provided: a normal approximation based interval, a quantile based interval and Fieller's confidence interval, all using perturbation-resampling. The user can also request an estimate of the incremental value of the surrogate marker information.

Usage

```
R.multiple.surv(xone, xzero, deltaone, deltazero, sone, szero, type = 1, t,
weight.perturb = NULL, landmark, extrapolate = FALSE, transform = FALSE,
conf.int = FALSE, var = FALSE, incremental.value = FALSE, approx = T)
```

Arguments

xone	numeric vector, the observed event times in the treatment group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
xzero	numeric vector, the observed event times in the control group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
deltaone	numeric vector, the event indicators for the treatment group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
deltazero	numeric vector, the event indicators for the control group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
sone	matrix of numeric values; surrogate marker measurements at t_0 for treated observations. If $X_{1i} < t_0$, then the surrogate marker measurements should be NA.
szero	matrix of numeric values; surrogate marker measurements at t_0 for control observations. If $X_{0i} < t_0$, then the surrogate marker measurements should be NA.
type	type of estimate; options are 1 = two-stage robust estimator, 2 = weighted two-stage robust estimator, 3 = double-robust estimator, 4 = two-stage model-based estimator, 5 = weighted estimator, 6 = double-robust model-based estimator; default is 1.
t	the time of interest.

<code>weight.perturb</code>	weights used for perturbation resampling.
<code>landmark</code>	the landmark time t_0 or time of surrogate marker measurement.
<code>extrapolate</code>	TRUE or FALSE; indicates whether the user wants to use extrapolation.
<code>transform</code>	TRUE or FALSE; indicates whether the user wants to use a transformation for the surrogate marker pseudo-score.
<code>conf.int</code>	TRUE or FALSE; indicates whether a 95% confidence interval for delta is requested, default is FALSE.
<code>var</code>	TRUE or FALSE; indicates whether a variance estimate is requested, default is FALSE.
<code>incremental.value</code>	TRUE or FALSE; indicates whether the user would like to see the incremental value of the surrogate marker information, default is FALSE.
<code>approx</code>	TRUE or FALSE indicating whether an approximation should be used when calculating the probability of censoring; most relevant in settings where the survival time of interest for the primary outcome is greater than the last observed event but before the last censored case, default is TRUE.

Details

Let $G \in \{A, B\}$ be the binary treatment indicator and we assume that subjects are randomly assigned to either treatment group A or B at baseline. Let T denote the time to the occurrence of the primary outcome, death for example, and $S = (S_1, S_2, \dots, S_k)^T$ denote the vector of k surrogate markers measured at a given time t_0 . Let $T^{(g)}$ and $S^{(g)}$ denote the counterfactual event time and surrogate marker measurements under treatment $G = g$ for $g \in \{A, B\}$. In practice, we only observe $(T, S) = (T^{(A)}, S^{(A)})$ or $(T^{(B)}, S^{(B)})$ depending on whether $G = A$ or B . The treatment effect, $\Delta(t)$, is the treatment difference in survival rates at time $t > t_0$, $\Delta(t) = E\{I(T^{(A)} > t)\} - E\{I(T^{(B)} > t)\} = P(T^{(A)} > t) - P(T^{(B)} > t)$ where $I(\cdot)$ is the indicator function. For individuals who are censored or experience the primary outcome before t_0 , we assume that their S information is not available.

The surrogate marker information at time t_0 is defined as a combination of the observed information on $I(T > t_0)$ and the observed S at t_0 , denoted by $Q_{t_0} = \{Q_{t_0}^{(g)}, g = A, B\}$, where $Q_{t_0}^{(g)} = \{T^{(g)} \wedge t_0, S^{(g)}I(T^{(g)} > t_0)\}$. With information on Q_{t_0} , the residual treatment effect is defined as: $\Delta_S(t, t_0) = E\{I(T^{(A)} > t) - I(T^{(B)} > t) \mid Q_{t_0}^{(A)} = Q_{t_0}^{(B)}\} = P(T^{(B)} > t_0) \int_S \psi_A(t \mid S, t_0) dF_B(S \mid t_0) - P(T^{(B)} > t)$ where $S = (s_1, \dots, s_k)^T$, $F_g(S \mid t_0) = P(S^{(g)} \leq S \mid T^{(g)} > t_0)$, $\psi_g(t \mid S, t_0) = P(T^{(g)} > t \mid S^{(g)} = S, T^{(g)} > t_0)$. The proportion of the treatment effect on the primary outcome that is explained by the treatment effect on Q_{t_0} is $R_S(t, t_0) = 1 - \Delta_S(t, t_0)/\Delta$. This function provides 6 different estimators for $R_S(t, t_0)$ using censored data.

Due to censoring, the observed data consist of n observations $\{(G_i, X_i, \delta_i, S_i I(X_i > t_0)), i = 1, \dots, n\}$ from the two treatment groups, where $X_i = \min(T_i, C_i)$, $\delta_i = I(T_i < C_i)$, and C_i denotes the censoring time for the i th subject. We assume independent censoring i.e., $(T_i, S_i) \perp C_i \mid G_i$. For ease of notation, we also let $\{(X_{gi}, \delta_{gi}, S_{gi} I(X_{gi} > t_0)), i = 1, \dots, n_g\}$ denote $n_g = \sum_{i=1}^n I(G_i = g)$ observations from treatment group $g \in \{A, B\}$, where $X_{gi} = \min(T_{gi}, C_{gi})$ and $\delta_{gi} = I(T_{gi} < C_{gi})$. Without loss of generality, we assume that $\bar{\pi}_g = n_g/n \rightarrow \pi_g \in (0, 1)$ as $n \rightarrow \infty$. Throughout, we estimate the treatment effect $\Delta(t) = P(T^{(A)} > t) - P(T^{(B)} > t)$ as $\hat{\Delta}(t) = n_A^{-1} \sum_{i=1}^{n_A} \frac{I(X_{Ai} > t)}{\hat{W}_A^C(t)} - n_B^{-1} \sum_{i=1}^{n_B} \frac{I(X_{Bi} > t)}{\hat{W}_B^C(t)}$ where $\hat{W}_g^C(t)$ is the Kaplan-Meier estimator of $W_g^C(t) = P(C_g > t)$.

We first describe the two-stage robust estimator which involves a two-stage procedure combining the use of a working model and a nonparametric estimation procedure for $\Delta_S(t, t_0)$. The idea is simply to summarize S into a univariate score U and then construct a nonparametric estimator for

$R_S(t, t_0)$ treating U as S . To construct U , we approximate the conditional distribution of $T^{(A)} | S^{(A)}, T^{(A)} > t_0$ by using a working semiparametric model such as the landmark proportional hazards model $q_A(S) = P(T^{(A)} > t | T^{(A)} > t_0, S^{(A)}) = \exp\{-\Lambda_0(t|t_0) \exp(\beta^T S^{(A)})\}, t > t_0$, where $\Lambda_0(t|t_0)$ is the unspecified baseline cumulative hazard function for $T^{(A)}$ conditional on $\{T^{(A)} > t_0\}$ and β is an unknown vector of coefficients. Let $\hat{\beta}$ be the maximizer of the corresponding log partial likelihood function and $\hat{\Lambda}_0(t|t_0)$ be the Breslow-type estimate of baseline hazard. If one were to assume that this working model is correctly specified, then a consistent estimate of $\Delta_S(t, t_0)$ would simply be: $\hat{\Delta}_S^M = n_B^{-1} \sum_{i=1}^{n_B} [\exp\{-\hat{\Lambda}_0(t|t_0) \exp(\hat{\beta}^T S_{Bi})\} \frac{I(X_{Bi} > t_0)}{\hat{W}_B^C(t_0)} - \frac{I(X_{Bi} > t)}{\hat{W}_B^C(t)}]$. We refer to this estimate as the two-stage model-based estimator (option 4 for type).

Instead of relying on correct specification of this model, we use the resulting score $U = \beta_0^T S$ as a univariate “pseudo-marker” to summarize the k surrogates. In the second stage, to estimate $\Delta_S(t, t_0)$, we apply a nonparametric approach with S represented by the univariate marker U . Specifically, we use a kernel Nelson-Aalen estimator to nonparametrically estimate $\hat{\phi}_A(t|u, t_0) = P(T^{(A)} > t | U^{(A)} = u, T^{(A)} > t_0) = \exp\{-\hat{\Lambda}_A(t|u, t_0)\}$ as $\hat{\phi}_A(t|u, t_0) = \exp\{-\hat{\Lambda}_A(t|u, t_0)\}$, where $\hat{\Lambda}_A(t|u, t_0) = \int_{t_0}^t \frac{\sum_{i=1}^{n_A} I(X_{Ai} > t_0) K_h\{\gamma(\hat{U}_{Ai}) - \gamma(u)\} dN_{Ai}(z)}{\sum_{i=1}^{n_A} I(X_{Ai} > t_0) K_h\{\gamma(\hat{U}_{Ai}) - \gamma(u)\} Y_{Ai}(z)}, \hat{U}_{Ai} = \hat{\beta}^T S_{Ai}, \hat{U}_{Bi} = \hat{\beta}^T S_{Bi}, Y_{Ai} = I(X_{Ai} \geq t), N_{Ai}(t) = I(X_{Ai} \leq t) \delta_{Ai}, K(\cdot)$ is a smooth symmetric density function, $K_h(x) = K(x/h)/h$, and $\gamma(\cdot)$ is a given monotone transformation function. We then estimate $\Delta_S(t, t_0)$ as $\hat{\Delta}_S(t, t_0) = n_B^{-1} \sum_{i=1}^{n_B} [\hat{\phi}_A(t|\hat{U}_{Bi}, t_0) \frac{I(X_{Bi} > t_0)}{\hat{W}_B^C(t_0)} - \frac{I(X_{Bi} > t)}{\hat{W}_B^C(t)}]$ and $\hat{R}_S(t, t_0) = 1 - \hat{\Delta}_S(t, t_0)/\hat{\Delta}(t)$. We refer to this estimate as the two-stage robust estimator (option 1 for type).

The next estimator borrows ideas from the extensive causal inference literature focusing on double robust estimators two-stage weighted estimator with a propensity score weight explicitly balancing the two treatment groups with respect to the distribution of S . The weighting enables us to “adjust” the distribution of $S^{(A)}$ before constructing the conditional survival estimate $\hat{\phi}_A(t|u, t_0)$. This approach results in a double-robust estimator of $\Delta_S(t, t_0)$, which is consistent when either $U^{(A)}$ captures all the information about the relationship between $I(T^{(A)} \geq t)$ and $S^{(A)}$ or the propensity score model for $\pi(S, t_0) = P(G_i = B | S_i = S, T_i > t_0)$ is correctly specified. While $\pi(S, t_0)$ depends on t_0 , for simplicity, we drop t_0 from our notation and simply use $\pi(S)$.

Regression models can be imposed to obtain estimates for $\pi(S)$. For example, a simple logistic regression model can be imposed for $\tilde{\pi}(S) = P(G_i = B | S_i = S, X_i > t_0)$ with $\text{logit}\{\tilde{\pi}(S)\} = \alpha_0 + \alpha_1^T S$, where α_0 and α_1 are estimated only among those with $X_{gi} > t_0$ to account for censoring. The propensity score of interest, $\pi(S)$, can be derived from $\tilde{\pi}(S)$ directly since $\text{logit}\{\pi(S)\} = \text{logit}\{\tilde{\pi}(S)\} + \log\{W_A^C(t_0)/W_B^C(t_0)\}$, which follows from the assumption that $(T_{gi}, S_{gi}) \perp C_{gi}$. We then modify the above expression by weighting observations with the estimated $L(S_{Ai}) = \pi(S_{Ai})/\{1 - \pi(S_{Ai})\}$ and obtain

$$\hat{\Lambda}_A^w(t|u, t_0) = \int_{t_0}^t \frac{\sum_{i=1}^{n_A} \hat{L}(S_{Ai}) I(X_{Ai} > t_0) K_h\{\gamma(\hat{U}_{Ai}) - \gamma(u)\} dN_{Ai}(z)}{\sum_{i=1}^{n_A} \hat{L}(S_{Ai}) I(X_{Ai} > t_0) K_h\{\gamma(\hat{U}_{Ai}) - \gamma(u)\} Y_{Ai}(z)},$$

, where $\hat{L}(S_{gi}) = \exp(\hat{\alpha}_0 + \hat{\alpha}_1^T S_{gi}) \hat{W}_B^C(t_0) / \hat{W}_A^C(t_0)$.

Subsequently, we define $\hat{\Delta}_S^w(t, t_0) = n_B^{-1} \sum_{i=1}^{n_B} [\hat{\phi}_A^w(t|\hat{U}_{Bi}, t_0) \frac{I(X_{Bi} > t_0)}{\hat{W}_B^C(t_0)} - \frac{I(X_{Bi} > t)}{\hat{W}_B^C(t)}]$ and $\hat{R}_S^w(t, t_0) = 1 - \hat{\Delta}_S^w(t, t_0)/\hat{\Delta}(t)$ where $\hat{\phi}_A^w(t|t_0, u) = \exp\{-\hat{\Lambda}_A^w(t|t_0, u)\}$. We refer to this estimate as the weighted two-stage robust estimator (option 2 for type).

While the two-stage weighted estimator reflects one way to enhance the robustness of an initial estimator, the idea of combining a propensity-score type model and a regression-type model has certainly been extensively studied in the causal inference literature and a more familiar double-robust estimator can be constructed as: $\hat{\Delta}_S^{DR}(t, t_0) = n^{-1} [\sum_{i=1}^{n_A} \frac{I(X_{Ai} > t)}{\hat{W}_A^C(t) \pi_B} \hat{L}(S_{Ai}) - \sum_{i=1}^{n_B} \frac{I(X_{Bi} > t)}{\hat{W}_B^C(t) \pi_B}] - n^{-1} [\sum_{i=1}^{n_A} \frac{\hat{\phi}_A(t|\hat{U}_{Ai}, t_0) I(X_{Ai} > t_0)}{\hat{W}_A^C(t_0) \pi_B} \hat{L}(S_{Ai}) - \sum_{i=1}^{n_B} \frac{\hat{\phi}_A(t|\hat{U}_{Bi}, t_0) I(X_{Bi} > t_0)}{\hat{W}_B^C(t_0) \pi_B}]$ and $\hat{R}_S^{DR}(t, t_0) = 1 -$

$\hat{\Delta}_S^{DR}(t, t_0)/\hat{\Delta}(t)$, where $\hat{\phi}_A(t|\hat{U}_{gi}, t_0)$ is the (unweighted) estimate of $\phi_A(t | u, t_0)$ used in $\hat{\Delta}_S(t, t_0)$. We refer to this estimate as the double robust estimator (option 3 for type).

The weighted estimator (option type 5) is defined as: $\hat{\Delta}_S^{PS}(t, t_0) = n^{-1} \sum_{i=1}^n \{ \frac{I(X_i > t)}{\hat{W}_{G_i}^C(t) \pi_B} [I(G_i = A) \hat{L}(S_{Ai}) - I(G_i = B)] \}$ and $\hat{R}_S^{PS}(t, t_0) = 1 - \hat{\Delta}_S^{PS}(t, t_0)/\hat{\Delta}(t)$. This estimator completely relies on the correct specification of $\pi(S)$. The double-robust model-based estimator (option 6 for type) is defined as $\hat{\Delta}_S^{DR2}(t, t_0)$ and $\hat{R}_S^{DR2}(t, t_0) = 1 - \hat{\Delta}_S^{DR2}(t, t_0)/\hat{\Delta}(t)$ which are constructed parallel to the construction of $\hat{\Delta}_S^{DR}(t, t_0)$ i.e., a combination of $\hat{\Delta}_S^M(t, t_0)$ and $\hat{R}_S^{PS}(t, t_0)$.

Variance estimates are obtained using perturbation resampling. If a confidence interval is requested three versions are provided: a normal approximation based interval, a quantile based interval and Fieller's confidence interval, all using perturbation-resampling. An estimate of the incremental value of the surrogate marker information can also be requested; this essentially compared the proportion explained by the surrogate information vs. the proportion explained by T alone up to t_0 . Details can be found in Parast, L., Cai, T., & Tian, L. (2021). Evaluating multiple surrogate markers with censored data. *Biometrics*, 77(4), 1315-1327.

Value

A list is returned:

delta	the estimate, $\hat{\Delta}(t)$, described in delta.estimate documentation.
delta.s	the residual treatment effect estimate, $\hat{\Delta}_S(t, t_0)$.
R.s	the estimated proportion of treatment effect explained by the set of markers, $\hat{R}_S(t, t_0)$.
delta.var	the variance estimate of $\hat{\Delta}(t)$; if var = TRUE or conf.int = TRUE.
delta.s.var	the variance estimate of $\hat{\Delta}_S(t, t_0)$; if var = TRUE or conf.int = TRUE.
R.s.var	the variance estimate of $\hat{R}_S(t, t_0)$; if var = TRUE or conf.int = TRUE.
conf.int.normal.delta	a vector of size 2; the 95% confidence interval for $\hat{\Delta}(t)$ based on a normal approximation; if conf.int = TRUE.
conf.int.quantile.delta	a vector of size 2; the 95% confidence interval for $\hat{\Delta}(t)$ based on sample quantiles of the perturbed values; if conf.int = TRUE.
conf.int.normal.delta.s	a vector of size 2; the 95% confidence interval for $\hat{\Delta}_S(t, t_0)$ based on a normal approximation; if conf.int = TRUE.
conf.int.quantile.delta.s	a vector of size 2; the 95% confidence interval for $\hat{\Delta}_S(t, t_0)$ based on sample quantiles of the perturbed values; if conf.int = TRUE.
conf.int.normal.R.s	a vector of size 2; the 95% confidence interval for $\hat{R}_S(t, t_0)$ based on a normal approximation; if conf.int = TRUE.
conf.int.quantile.R.s	a vector of size 2; the 95% confidence interval for $\hat{R}_S(t, t_0)$ based on sample quantiles of the perturbed values; if conf.int = TRUE.
conf.int.fieller.R.s	a vector of size 2; the 95% confidence interval for $\hat{R}_S(t, t_0)$ based on Fieller's approach; if conf.int = TRUE.
delta.t	the estimate, $\hat{\Delta}_T(t, t_0)$; if incremental.vaue = TRUE.

R.t	the estimated proportion of treatment effect explained by survival only, $\hat{R}_T(t, t_0)$; if incremental.vaue = TRUE.
incremental.value	the estimate of the incremental value of the surrogate markers, $\hat{IV}_S(t, t_0)$; if incremental.vaue = TRUE.
delta.t.var	the variance estimate of $\hat{\Delta}_T(t, t_0)$; if var = TRUE or conf.int = TRUE and incremental.vaue = TRUE.
R.t.var	the variance estimate of $\hat{R}_T(t, t_0)$; if var = TRUE or conf.int = TRUE and incremental.vaue = TRUE.
incremental.value.var	the variance estimate of $\hat{IV}_S(t, t_0)$; if var = TRUE or conf.int = TRUE and incremental.vaue = TRUE.
conf.int.normal.delta.t	a vector of size 2; the 95% confidence interval for $\hat{\Delta}_T(t, t_0)$ based on a normal approximation; if conf.int = TRUE and incremental.vaue = TRUE.
conf.int.quantile.delta.t	a vector of size 2; the 95% confidence interval for $\hat{\Delta}_T(t, t_0)$ based on sample quantiles of the perturbed values; if conf.int = TRUE and incremental.vaue = TRUE.
conf.int.normal.R.t	a vector of size 2; the 95% confidence interval for $\hat{R}_T(t, t_0)$ based on a normal approximation; if conf.int = TRUE and incremental.vaue = TRUE.
conf.int.quantile.R.t	a vector of size 2; the 95% confidence interval for $\hat{R}_T(t, t_0)$ based on sample quantiles of the perturbed values; if conf.int = TRUE and incremental.vaue = TRUE.
conf.int.fieller.R.t	a vector of size 2; the 95% confidence interval for $\hat{R}_T(t, t_0)$ based on Fieller's approach; if conf.int = TRUE and incremental.vaue = TRUE.
conf.int.normal.iv	a vector of size 2; the 95% confidence interval for $\hat{IV}_S(t, t_0)$ based on a normal approximation; if conf.int = TRUE and incremental.vaue = TRUE.
conf.int.quantile.iv	a vector of size 2; the 95% confidence interval for $\hat{IV}_S(t, t_0)$ based on sample quantiles of the perturbed values; if conf.int = TRUE and incremental.vaue = TRUE.

Note

If the treatment effect is not significant, the user will receive the following message: "Warning: it looks like the treatment effect is not significant; may be difficult to interpret the residual treatment effect in this setting".

Author(s)

Layla Parast

References

- Parast, L., Cai, T., & Tian, L. (2017). Evaluating surrogate marker information using censored data. *Statistics in Medicine*, 36(11), 1767-1782.
- Parast, L., Cai, T., & Tian, L. (2021). Evaluating multiple surrogate markers with censored data. *Biometrics*, 77(4), 1315-1327.

Examples

```
data(d_example_multiple)
names(d_example_multiple)
## Not run:
R.multiple.surv(xone = d_example_multiple$x1, xzero = d_example_multiple$x0, deltaone =
d_example_multiple$delta1, deltazero = d_example_multiple$delta0, sone =
as.matrix(d_example_multiple$s1), szero = as.matrix(d_example_multiple$s0),
type =1, t = 1, landmark=0.5)
R.multiple.surv(xone = d_example_multiple$x1, xzero = d_example_multiple$x0, deltaone =
d_example_multiple$delta1, deltazero = d_example_multiple$delta0, sone =
as.matrix(d_example_multiple$s1), szero = as.matrix(d_example_multiple$s0),
type =1, t = 1, landmark=0.5, conf.int = T)
R.multiple.surv(xone = d_example_multiple$x1, xzero = d_example_multiple$x0, deltaone =
d_example_multiple$delta1, deltazero = d_example_multiple$delta0, sone =
as.matrix(d_example_multiple$s1), szero = as.matrix(d_example_multiple$s0),
type =3, t = 1, landmark=0.5)

## End(Not run)
```

R.s.estimate

Calculates the proportion of treatment effect explained

Description

This function calculates the proportion of treatment effect on the primary outcome explained by the treatment effect on the surrogate marker(s). This function is intended to be used for a fully observed continuous outcome. The user can also request a variance estimate and a 95% confidence interval, both estimated using perturbing-resampling. If a confidence interval is requested three versions are provided: a normal approximation based interval, a quantile based interval, and Fieller's confidence interval.

Usage

```
R.s.estimate(sone, szero, yone, yzero, var = FALSE, conf.int = FALSE,
weight.perturb = NULL, number = "single", type = "robust", extrapolate = FALSE,
transform = FALSE, warn.te = FALSE, warn.support = FALSE)
```

Arguments

sone	numeric vector or matrix; surrogate marker for treated observations, assumed to be continuous. If there are multiple surrogates then this should be a matrix with n_1 (number of treated observations) rows and n.s (number of surrogate markers) columns.
szero	numeric vector; surrogate marker for control observations, assumed to be continuous. If there are multiple surrogates then this should be a matrix with n_0 (number of control observations) rows and n.s (number of surrogate markers) columns.
yone	numeric vector; primary outcome for treated observations, assumed to be continuous.
yzero	numeric vector; primary outcome for control observations, assumed to be continuous.

var	TRUE or FALSE; indicates whether a variance estimate is requested, default is FALSE.
conf.int	TRUE or FALSE; indicates whether a 95% confidence interval is requested, default is FALSE
weight.perturb	a $n_1 + n_0$ by x matrix of weights where n_1 = length of yone and n_0 = length of yzero; used for perturbation-resampling, default is null.
number	specifies the number of surrogate markers; choices are "multiple" or "single", default is "single"
type	specifies the type of estimation; choices are "robust" or "model" or "freedman", default is "robust"
extrapolate	TRUE or FALSE; indicates whether the user wants to use extrapolation.
transform	TRUE or FALSE; indicates whether the user wants to use a transformation for the surrogate marker.
warn.te	value to control warnings; user does not need to specify.
warn.support	value to control warnings; user does not need to specify.

Details

Let $Y^{(1)}$ and $Y^{(0)}$ denote the primary outcome under the treatment and primary outcome under the control, respectively. Let $S^{(1)}$ and $S^{(0)}$ denote the surrogate marker under the treatment and the surrogate marker under the control, respectively. The residual treatment effect is defined as

$$\Delta_S = \int_{-\infty}^{\infty} E(Y^{(1)}|S^{(1)} = s) dF_0(s) - \int_{-\infty}^{\infty} E(Y^{(0)}|S^{(0)} = s) dF_0(s),$$

where $\Delta_S(s) = E(Y^{(1)}|S^{(1)} = s) - E(Y^{(0)}|S^{(0)} = s)$ and $F_0(\cdot)$ is the marginal cumulative distribution function of $S^{(0)}$, the surrogate marker measure under the control. The proportion of treatment effect explained by the surrogate marker, which we denote by R_S , can be expressed using a contrast between Δ_S and Δ :

$$R_S = \{\Delta - \Delta_S\} / \Delta = 1 - \Delta_S / \Delta.$$

The definition and estimation of Δ is described in the delta.estimate documentation.

A flexible model-based approach to estimate Δ_S in the single marker setting is to specify:

$$E(S^{(0)}) = \alpha_0 \quad \text{and} \quad E(S^{(1)}) - E(S^{(0)}) = \alpha_1,$$

$$E(Y^{(0)}|S^{(0)}) = \beta_0 + \beta_1 S^{(0)} \quad \text{and} \quad E(Y^{(1)}|S^{(1)}) = (\beta_0 + \beta_2) + (\beta_1 + \beta_3) S^{(1)}.$$

It can be shown that when these models hold, $\Delta_S = \beta_2 + \beta_3 \alpha_0$. Thus, reasonable estimates for Δ_S and R_S using this approach would be $\hat{\Delta}_S = \hat{\beta}_2 + \hat{\beta}_3 \hat{\alpha}_0$ and $\hat{R}_S = 1 - \hat{\Delta}_S / \hat{\Delta}$.

For robust estimation of Δ_S in the single marker setting, we estimate $\mu_1(s) = E(Y^{(1)}|S^{(1)} = s)$ nonparametrically using kernel smoothing:

$$\hat{\mu}_1(s) = \frac{\sum_{i=1}^{n_1} K_h(S_{1i} - s) Y_{1i}}{\sum_{i=1}^{n_1} K_h(S_{1i} - s)}$$

where S_{1i} is the observed $S^{(1)}$ for person i , Y_{1i} is the observed $Y^{(1)}$ for person i , $K(\cdot)$ is a smooth symmetric density function with finite support, $K_h(\cdot) = K(\cdot/h)/h$ and h is a specified bandwidth. As in most nonparametric functional estimation procedures, the choice of the smoothing parameter h is critical. To eliminate the impact of the bias of the conditional mean function on the resulting estimator, we require the standard undersmoothing assumption of $h = O(n_1^{-\delta})$ with $\delta \in (1/4, 1/3)$.

To obtain an appropriate h we first use `bw.nrd` to obtain h_{opt} ; and then we let $h = h_{opt}n_1^{-c_0}$ with $c_0 = 0.25$. We then estimate Δ_S as

$$\hat{\Delta}_S = \sum_{i=1}^{n_0} \frac{\hat{\mu}_1(S_{0i}) - Y_{0i}}{n_0}$$

where S_{0i} is the observed $S^{(0)}$ for person i and Y_{0i} is the observed $Y^{(0)}$ for person i . Lastly, we estimate R_S as $\hat{R}_S = 1 - \hat{\Delta}_S / \hat{\Delta}$.

This function also allows for estimation of R_S using Freedman's approach. Let Y denote the primary outcome, S denote the surrogate marker, and G denote the treatment group (0 for control, 1 for treatment). Freedman's approach to calculating the proportion of treatment effect explained by the surrogate marker is to fit the following two regression models:

$$E(Y|G) = \gamma_0 + \gamma_1 I(G = 1) \quad \text{and} \quad E(Y|G, S) = \gamma_{0S} + \gamma_{1S} I(G = 1) + \gamma_{2S} S$$

and estimating the proportion of treatment effect explained, denoted by R_S , as $1 - \hat{\gamma}_{1S} / \hat{\gamma}_1$.

This function also estimates R_S in a multiple marking setting. A flexible model-based approach to estimate Δ_S in the multiple marker setting is to specify models for $E(Y|G, S)$ and $E(S_j|G)$ for each S_j in $S = \{S_1, \dots, S_p\}$ (where p is the number of surrogate markers). Without loss of generality, consider the case where there are three surrogate markers, $S = \{S_1, S_2, S_3\}$ and one specifies the following linear models:

$$\begin{aligned} E(Y^{(0)}|S^{(0)}) &= \beta_0 + \beta_1 S_1^{(0)} + \beta_2 S_2^{(0)} + \beta_3 S_3^{(0)} \\ E(Y^{(1)}|S^{(1)}) &= (\beta_0 + \beta_4) + (\beta_1 + \beta_5) S_1^{(1)} + (\beta_2 + \beta_6) S_2^{(1)} + (\beta_3 + \beta_7) S_3^{(1)} \\ E(S_j^{(0)}) &= \alpha_j, \quad j = 1, 2, 3. \end{aligned}$$

It can be shown that when these models hold

$$\Delta_S = \beta_4 + \beta_5 \alpha_1 + \beta_6 \alpha_2 + \beta_7 \alpha_3.$$

Thus, reasonable estimates for Δ_S and R_S here would be easily obtained by replacing the unknown regression coefficients in the models above by their consistent estimators.

For robust estimation of Δ_S in the multiple marker setting, we use a two-stage procedure combining the model-based approach and the nonparametric estimation procedure from the single marker setting. Specifically, we use a working semiparametric model:

$$E(Y^{(1)}|S^{(1)} = S) = \beta_0 + \beta_1 S_1^{(1)} + \beta_2 S_2^{(1)} + \beta_3 S_3^{(1)}$$

and define $Q^{(1)} = \hat{\beta}_0 + \hat{\beta}_1 S_1^{(1)} + \hat{\beta}_2 S_2^{(1)} + \hat{\beta}_3 S_3^{(1)}$ and $Q^{(0)} = \hat{\beta}_0 + \hat{\beta}_1 S_1^{(0)} + \hat{\beta}_2 S_2^{(0)} + \hat{\beta}_3 S_3^{(0)}$ to reduce the dimension of S in the first stage and in the second stage, we apply the robust approach used in the single marker setting to estimate its surrogacy.

To use Freedman's approach in the presence of multiple markers, the markers are simply additively entered into the second regression model.

Variance estimation and confidence interval construction are performed using perturbation-resampling. Specifically, let $\{V^{(b)} = (V_{11}^{(b)}, \dots, V_{1n_1}^{(b)}, V_{01}^{(b)}, \dots, V_{0n_0}^{(b)})^T, b = 1, \dots, D\}$ be $n \times D$ independent copies of a positive random variables V from a known distribution with unit mean and unit variance. Let

$$\hat{\Delta}^{(b)} = \frac{\sum_{i=1}^{n_1} V_{1i}^{(b)} Y_{1i}}{\sum_{i=1}^{n_1} V_{1i}^{(b)}} - \frac{\sum_{i=1}^{n_0} V_{0i}^{(b)} Y_{0i}}{\sum_{i=1}^{n_0} V_{0i}^{(b)}}.$$

The variance of $\hat{\Delta}$ is obtained as the empirical variance of $\{\hat{\Delta}^{(b)}, b = 1, \dots, D\}$. In this package, we use weights generated from an Exponential(1) distribution and use $D = 500$. Variance estimates for $\hat{\Delta}_S$ and $\hat{R}_S$ are calculated similarly. We construct two versions of the 95% confidence interval for each estimate: one based on a normal approximation confidence interval using the estimated variance and another taking the 2.5th and 97.5th empirical percentile of the perturbed quantities. In addition, we use Fieller's method to obtain a third confidence interval for R_S as

$$\left\{ 1 - r : \frac{(\hat{\Delta}_S - r\hat{\Delta})^2}{\hat{\sigma}_{11} - 2r\hat{\sigma}_{12} + r^2\hat{\sigma}_{22}} \leq c_\alpha \right\},$$

where $\hat{\Sigma} = (\hat{\sigma}_{ij})_{1 \leq i, j \leq 2}$ and c_α is the $(1 - \alpha)$ th percentile of

$$\left\{ \frac{\{\hat{\Delta}_S^{(b)} - (1 - \hat{R}_S)\hat{\Delta}^{(b)}\}^2}{\hat{\sigma}_{11} - 2(1 - \hat{R}_S)\hat{\sigma}_{12} + (1 - \hat{R}_S)^2\hat{\sigma}_{22}}, b = 1, \dots, C \right\}$$

where $\alpha = 0.05$.

Note that if the observed supports for S are not the same, then $\hat{\mu}_1(s)$ for $S_{0i} = s$ outside the support of S_{1i} may return NA (depending on the bandwidth). If `extrapolation = TRUE`, then the $\hat{\mu}_1(s)$ values for these surrogate values are set to the closest non-NA value. If `transform = TRUE`, then S_{1i} and S_{0i} are transformed such that the new transformed values, S_{1i}^{tr} and S_{0i}^{tr} are defined as: $S_{gi}^{tr} = F([S_{gi} - \mu]/\sigma)$ for $g = 0, 1$ where $F(\cdot)$ is the cumulative distribution function for a standard normal random variable, and μ and σ are the sample mean and standard deviation, respectively, of $(S_{1i}, S_{0i})^T$.

Value

A list is returned:

<code>R.s</code>	the estimate, $\hat{R}_S$, described above.
<code>R.s.var</code>	the variance estimate of $\hat{R}_S$; if <code>var = TRUE</code> or <code>conf.int = TRUE</code> .
<code>conf.int.normal.R.s</code>	a vector of size 2; the 95% confidence interval for $\hat{R}_S$ based on a normal approximation; if <code>conf.int = TRUE</code> .
<code>conf.int.quantile.R.s</code>	a vector of size 2; the 95% confidence interval for $\hat{R}_S$ based on sample quantiles of the perturbed values, described above; if <code>conf.int = TRUE</code> .
<code>conf.int.fieller.R.s</code>	a vector of size 2; the 95% confidence interval for $\hat{R}_S$ based on Fieller's approach, described above; if <code>conf.int = TRUE</code> .

For all options other than "freedman", the following are also returned:

<code>delta</code>	the estimate, $\hat{\Delta}$, described in <code>delta.estimate</code> documentation.
<code>delta.s</code>	the estimate, $\hat{\Delta}_S$, described above.
<code>delta.var</code>	the variance estimate of $\hat{\Delta}$; if <code>var = TRUE</code> or <code>conf.int = TRUE</code> .
<code>delta.s.var</code>	the variance estimate of $\hat{\Delta}_S$; if <code>var = TRUE</code> or <code>conf.int = TRUE</code> .
<code>conf.int.normal.delta</code>	a vector of size 2; the 95% confidence interval for $\hat{\Delta}$ based on a normal approximation; if <code>conf.int = TRUE</code> .
<code>conf.int.quantile.delta</code>	a vector of size 2; the 95% confidence interval for $\hat{\Delta}$ based on sample quantiles of the perturbed values, described above; if <code>conf.int = TRUE</code> .

`conf.int.normal.delta.s`

a vector of size 2; the 95% confidence interval for $\hat{\Delta}_S$ based on a normal approximation; if `conf.int = TRUE`.

`conf.int.quantile.delta.s`

a vector of size 2; the 95% confidence interval for $\hat{\Delta}_S$ based on sample quantiles of the perturbed values, described above; if `conf.int = TRUE`.

Note

If the treatment effect is not significant, the user will receive the following message: "Warning: it looks like the treatment effect is not significant; may be difficult to interpret the proportion of treatment effect explained in this setting". In the single marker case with the robust estimation approach, if the observed support of the surrogate marker for the control group is outside the observed support of the surrogate marker for the treatment group, the user will receive the following message: "Warning: observed supports do not appear equal, may need to consider a transformation or extrapolation"

Author(s)

Layla Parast

References

- Freedman, L. S., Graubard, B. I., & Schatzkin, A. (1992). Statistical validation of intermediate endpoints for chronic diseases. *Statistics in medicine*, 11(2), 167-178.
- Parast, L., McDermott, M., Tian, L. (2016). Robust estimation of the proportion of treatment effect explained by surrogate marker information. *Statistics in Medicine*, 35(10):1637-1653.
- Wang, Y., & Taylor, J. M. (2002). A measure of the proportion of treatment effect explained by a surrogate marker. *Biometrics*, 58(4), 803-812.
- Fieller, Edgar C. (1954). Some problems in interval estimation. *Journal of the Royal Statistical Society. Series B (Methodological)*, 175-185.
- Fieller, E. C. (1940). The biological standardization of insulin. *Supplement to the Journal of the Royal Statistical Society*, 1-64.

Examples

```
data(d_example)
names(d_example)
R.s.estimate(yone=d_example$y1, yzero=d_example$y0, sone=d_example$s1.a, szero=d_example$s0.a,
number = "single", type = "robust")
R.s.estimate(yone=d_example$y1, yzero=d_example$y0, sone=cbind(d_example$s1.a,d_example$s1.b,
d_example$s1.c), szero=cbind(d_example$s0.a, d_example$s0.b, d_example$s0.c),
number = "multiple", type = "model")
```

R.s.estimate.me

Calculates the proportion of treatment effect explained correcting for measurement error in the surrogate marker

Description

This function calculates the proportion of treatment effect on the primary outcome explained by the treatment effect on a surrogate marker, correcting for measurement error in the surrogate marker. This function is intended to be used for a fully observed continuous outcome. The user must specify what type of estimation they would like (parametric or nonparametric estimation of the proportion explained, denoted by R) and what estimator they would like (see below for details).

Usage

```
R.s.estimate.me(sone, szero, yone, yzero, parametric = FALSE, estimator = "n",
me.variance, extrapolate = TRUE, transform = FALSE, naive = FALSE, Ronly = TRUE)
```

Arguments

sone	numeric vector or matrix; surrogate marker for treated observations, assumed to be continuous. If there are multiple surrogates then this should be a matrix with n_1 (number of treated observations) rows and n.s (number of surrogate markers) columns.
szero	numeric vector; surrogate marker for control observations, assumed to be continuous. If there are multiple surrogates then this should be a matrix with n_0 (number of control observations) rows and n.s (number of surrogate markers) columns.
yone	numeric vector; primary outcome for treated observations, assumed to be continuous.
yzero	numeric vector; primary outcome for control observations, assumed to be continuous.
parametric	TRUE or FALSE; indicates whether the user wants the parametric approach to be used (TRUE) or nonparametric (FALSE).
estimator	options are "d", "q", "n" for parametric and "q", "n" for nonparametric; "d" stands for the disattenuated estimator, "q" stands for the SIMEX estimator with quadratic extrapolation, "n" stands for the SIMEX estimator with a nonlinear extrapolation. Note that the nonlinear extrapolation may have convergence issues with a small sample size; if this occurs, please consider using quadratic extrapolation instead.
me.variance	the variance of the measurement error; must be provided.
extrapolate	TRUE or FALSE; indicates whether the user wants to use extrapolation.
transform	TRUE or FALSE; indicates whether the user wants to use a transformation for the surrogate marker.
naive	TRUE or FALSE; indicates whether the user wants the naive estimate (not correcting for measurement error) to also be calculated
Ronly	TRUE or FALSE; indicates whether the user wants only R (and corresponding variance and confidence intervals) to be returned.

Details

While there are many methods available to quantify the value of a surrogate marker, most assume that the marker is measured without error. This function calculates the proportion of treatment effect on the primary outcome explained by the treatment effect on a surrogate marker, correcting for measurement error in the surrogate marker. The user can choose either the parametric framework

or nonparametric framework for estimation. Within the parametric framework there are three options for measurement error correction: the disattenuated estimator, the SIMEX estimator with quadratic extrapolation, and the SIMEX estimator with nonlinear extrapolation. Within the nonparametric framework there are two options for measurement error correction: the SIMEX estimator with quadratic extrapolation and the SIMEX estimator with nonlinear extrapolation. We describe each below.

Let G be the binary treatment indicator with $G = 1$ indicating treatment and $G = 0$ indicating control (or placebo). We assume throughout that subjects are randomly assigned to treatment or control at baseline. Let Y and S denote the continuous primary outcome and continuous surrogate marker, respectively, where S is measured post-baseline and is assumed to be a biomarker, clinical measurement, psychological test score, or other physiological measurement. In the absence of measurement error, the observed data consists of $\{Y_i, S_i, G_i\}$ for $i \in \{1, \dots, n\}$. With measurement error, instead of observing S we observe $W = S + U$, where $E(U|S) = 0$ and the variance of U is σ_u^2 . Such measurement error may be attributable to, for example, laboratory error. Thus, our observed data will consist of $\{Y_i, W_i, G_i\}$ for $i \in \{1, \dots, n\}$. Throughout, we assume that σ_u^2 is known. Here, we are interested in estimating the proportion of the treatment effect on the primary outcome that is explained by the treatment effect on the surrogate marker, denoted as R_S .

To estimate R_S parametrically, we assume the following models $E(Y|G) = \beta_0 + \beta_1 G$ and $E(Y|G, S) = \beta_0^* + \beta_1^* G + \beta_2^* S$. It can be shown that if these models hold, $R_S = 1 - \beta_1^*/\beta_1$. When $W = S + U$ is available instead of S , this measurement error does not affect estimation of β_1 , but it does affect estimation of β_1^* , and β_2^* . Since estimation of R_S relies on estimation of β_1 and β_1^* , we focus on the effect of measurement error on β_1^* estimation. The attenuation bias for $\hat{\beta}_1^*$ and $\hat{R}$ can be written out in closed form when the proportion of treatment effect is parametrically estimated as described above, when these specified models hold, and when the surrogate marker S is measured with error. There exist two methods to eliminate this bias when estimating R_S . Taking advantage of the fact that we can express the attenuation bias in closed form, the first is a straightforward disattenuated estimator: $\hat{\beta}_{1A} = \hat{\beta}_1^* - \frac{\hat{\beta}_2^* \{\Omega_W^2 \Omega_{GW} - \Omega_{GW} (\Omega_W^2 - \sigma_u^2)\}}{\Omega_G^2 (\Omega_W^2 - \sigma_u^2) - \Omega_{GW} \Omega_{GW}}$ and $\hat{R}_A = 1 - \left[\hat{\beta}_1^* - \frac{\hat{\beta}_2^* \{\Omega_W^2 \Omega_{GW} - \Omega_{GW} (\Omega_W^2 - \sigma_u^2)\}}{\Omega_G^2 (\Omega_W^2 - \sigma_u^2) - \Omega_{GW} \Omega_{GW}} \right] / \hat{\beta}_1$ where Ω^2 denotes the sample variance or covariance.

The second method to eliminate this bias uses Simulation Extrapolation (SIMEX) estimation, which is a simulation-based method that involves first generating additional measurement error and observing how it affects the bias of the parameter estimate of interest, and then extrapolating this information to a setting with no measurement error. To incorporate SIMEX estimation within our surrogate marker framework, we define $W_{b,i}(\lambda) = W_i + \lambda^{1/2} \sigma_u \epsilon_{i,b}$ for $b = 1, \dots, B$ where $B = 50$, $\epsilon_{i,b} \sim N(0, 1)$, σ_u is assumed known, and $\lambda \in (0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0)$ and for each iteration b and λ value, obtaining $\hat{\beta}_{1b}^*(\lambda)$ by fitting the regression model: $E(Y | W_b(\lambda), S) = \beta_{0b}^* + \beta_{1b}^* W_b(\lambda) + \beta_{2b}^* S$. We then calculate the average estimate for each quantity over the iterations $b = 1, \dots, B$ for each λ value, denoted as $\hat{\beta}_{1,S,\sigma_u^2(1+\lambda)}^* = \sum_{b=1}^B \hat{\beta}_{1b}^*(\lambda)$. The second step, extrapolation, takes these average estimates for each λ value and extrapolates using a function $G(\Gamma, \lambda)$ to obtain the estimated quantity if $\lambda = -1$. For the extrapolation step, we use both a quadratic extrapolation and nonlinear extrapolation i.e., we solve for $\Gamma = (\alpha_0, \alpha_1, \alpha_2)^T$ in $\hat{\beta}_{1,S,\sigma_u^2(1+\lambda)}^* = \alpha_0 + \alpha_1 \lambda + \alpha_2 \lambda^2$ and $\hat{\beta}_{1,S,\sigma_u^2(1+\lambda)}^* = \alpha_0 + \alpha_1 / (\alpha_2 + \lambda)$, respectively. Using the estimates of $\alpha_0, \alpha_1, \alpha_2$, we calculate the predicted $\hat{\beta}_{1,S,\sigma_u^2(1+\lambda)}^*$ when $\lambda = -1$. In essence, the simulations add successively larger measurement errors of size $(1 + \lambda)\sigma_u^2$ and then extrapolate to the case when $\lambda = -1$ such that the measurement error is 0. We denote the resulting estimator of β_1^* as $\hat{\beta}_{1,SIMEX}^* = G(\hat{\Gamma}, -1)$ and define $\hat{R}_{SIMEX} = 1 - \hat{\beta}_{1,SIMEX}^* / \hat{\beta}_1$.

While the parametric approach to estimate the proportion of treatment effect explained by S is most commonly used in clinical practice, previous work has demonstrated biased results when the assumed models are not correctly specified. An alternative approach involves estimating the treatment effect, Δ , and residual treatment effect, Δ_S , as R_S is defined as $1 - \Delta / \Delta_S$. The quantity Δ can be estimated simply by $\hat{\Delta} = n_1^{-1} \sum_{i=1}^n Y_i I(G_i = 1) - n_0^{-1} \sum_{i=1}^n Y_i I(G_i = 0)$, where n_1 and

n_0 denote the number of individuals in the treatment and control groups, respectively. The quantity Δ_S can be estimated nonparametrically using kernel smoothing as $\hat{\Delta}_S = n_0^{-1} \sum_{i:G_i=0} \hat{\mu}_1(S_i) - n_0^{-1} \sum_{i=1}^n Y_i I(G_i = 0)$ where $\hat{\mu}_1(s) = \{\sum_{j:G_j=1} K_h(S_j - s) Y_j\} / \{\sum_{j:G_j=1} K_h(S_j - s)\}$, $K(\cdot)$ is a smooth symmetric density function with finite support, $K_h(\cdot) = K(\cdot/h)/h$ and h is a specified bandwidth such that $h = O(n_1^{-\nu})$ with $\nu \in (1/4, 1/2)$.

When $W = S + U$ is available instead of S , estimation of Δ is not affected whereas estimation of Δ_S is affected and thus, the nonparametric estimation procedure described above results in a biased estimate of R_S . Unlike the parametric approach, the attenuation bias cannot be expressed in closed form. Within this nonparametric framework, SIMEX estimation can be used to correct for measurement error. We implement the estimation procedure as described above where we first generate additional measurement error to obtain $W_{b,i}(\lambda)$ and for each iteration b and λ values obtain $\hat{\Delta}_{S,b}(\lambda) = n_0^{-1} \sum_{i:G_i=0} \left\{ \frac{\sum_{j:G_j=1} K_h(W_{b,j}(\lambda) - W_{b,i}(\lambda)) Y_j}{\sum_{j:G_j=1} K_h(W_{b,j}(\lambda) - W_{b,i}(\lambda))} \right\} - n_0^{-1} \sum_{i=1}^n Y_i I(G_i = 0)$. We then calculate the average estimate for each quantity over the iterations $b = 1, \dots, B$ for each λ value, denoted as $\hat{\Delta}_{S,\sigma_u^2(1+\lambda)} = \sum_{b=1}^B \hat{\Delta}_{S,b}(\lambda)$ and extrapolate using a function $G(\Gamma, \lambda)$; we specifically use the quadratic and nonlinear functions as in the parametric setting. We denote the resulting estimator of Δ_S as $\hat{\Delta}_{S,SIMEX} = G(\hat{\Gamma}, -1)$ and define $\hat{R}_{S,SIMEX} = 1 - \hat{\Delta}_{S,SIMEX} / \hat{\Delta}$.

In this function, parametric estimation is equivalent to Freedman's approach in the R.s.estimate documentation; nonparametric estimation is equivalent to the robust approach in the R.s.estimate documentation. Variance estimates for all estimators are calculated in this function based on derived closed form variance expressions. For all approaches, confidence intervals for Δ_S can be constructed using a normal approximation; confidence intervals for R_S can be constructed using either a normal approximation or using Fieller's method, all of which are provided in this function. Details regarding the asymptotic properties of these estimators and closed form variance calculation can be found in: Parast, L., Garcia, T. P., Prentice, R. L., & Carroll, R. J. (2022). Robust methods to correct for measurement error when evaluating a surrogate marker. *Biometrics*, 78(1), 9-23.

Value

A list is returned:

R.naive	the naive estimate of the proportion of treatment effect explained by the surrogate marker; only if naive = TRUE
R.naive.var	the estimated variance of the naive estimate of the proportion of treatment effect explained by the surrogate marker; only if naive = TRUE
R.naive.CI.normal	the 95% confidence interval using the normal approximation for the naive estimate of the proportion of treatment effect explained by the surrogate marker; only if naive = TRUE
R.naive.CI.fieller	the 95% confidence interval using Fieller's approach for the naive estimate of the proportion of treatment effect explained by the surrogate marker; only if naive = TRUE
B1star.naive	the naive estimate of the adjusted regression coefficient for treatment; only if naive = TRUE and Ronly = FALSE and parametric = TRUE
B1star.naive.var	the estimated variance of the naive estimate of the adjusted regression coefficient for treatment; only if naive = TRUE and Ronly = FALSE and parametric = TRUE

<code>B1star.naive.CI.normal</code>	the 95% confidence interval using the normal approximation for the naive estimate of the adjusted regression coefficient for treatment; only if <code>naive = TRUE</code> and <code>Ronly = FALSE</code> and <code>parametric = TRUE</code>
<code>deltas.naive</code>	the naive estimate of the residual treatment effect; only if <code>naive = TRUE</code> and <code>Ronly = FALSE</code> and <code>parametric = FALSE</code>
<code>deltas.naive.var</code>	the estimated variance of the naive estimate of the residual treatment effect; only if <code>naive = TRUE</code> and <code>Ronly = FALSE</code> and <code>parametric = FALSE</code>
<code>deltas.naive.CI.normal</code>	the 95% confidence interval using the normal approximation for the naive estimate of the residual treatment effect; only if <code>naive = TRUE</code> and <code>Ronly = FALSE</code> and <code>parametric = FALSE</code>
<code>R.corrected.dis</code>	the corrected disattenuated estimate of the proportion of treatment effect explained by the surrogate marker; only if <code>parametric = TRUE</code> and <code>estimator = "d"</code>
<code>R.corrected.var.dis</code>	the estimated variance of the corrected disattenuated estimate of the proportion of treatment effect explained by the surrogate marker; only if <code>naive = TRUE</code>
<code>R.corrected.CI.normal.dis</code>	the 95% confidence interval using the normal approximation for the corrected disattenuated estimate of the proportion of treatment effect explained by the surrogate marker; only if <code>parametric = TRUE</code> and <code>estimator = "d"</code>
<code>R.corrected.CI.fieller.dis</code>	the 95% confidence interval using Fieller's approach for the corrected disattenuated estimate of the proportion of treatment effect explained by the surrogate marker; only if <code>parametric = TRUE</code> and <code>estimator = "d"</code>
<code>B1star.corrected.dis</code>	the corrected disattenuated estimate of the adjusted regression coefficient for treatment; only if <code>parametric = TRUE</code> and <code>estimator = "d"</code> and <code>Ronly = FALSE</code>
<code>B1star.corrected.var.dis</code>	the estimated variance of the corrected disattenuated estimate of the adjusted regression coefficient for treatment; only if <code>parametric = TRUE</code> and <code>estimator = "d"</code> and <code>Ronly = FALSE</code>
<code>B1star.corrected.CI.normal.dis</code>	the 95% confidence interval using the normal approximation for the corrected disattenuated estimate of the adjusted regression coefficient for treatment; only if <code>parametric = TRUE</code> and <code>estimator = "d"</code> and <code>Ronly = FALSE</code>
<code>R.corrected.q</code>	the corrected SIMEX (quadratic) estimate of the proportion of treatment effect explained by the surrogate marker; only if <code>estimator = "q"</code>
<code>R.corrected.var.q</code>	the estimated variance of the corrected SIMEX (quadratic) estimate of the proportion of treatment effect explained by the surrogate marker; only if <code>estimator = "q"</code>
<code>R.corrected.CI.normal.q</code>	the 95% confidence interval using the normal approximation for the corrected SIMEX (quadratic) estimate of the proportion of treatment effect explained by the surrogate marker; only if <code>estimator = "q"</code>

`R.corrected.CI.fieller.q`
the 95% confidence interval using Fieller's approach for the corrected SIMEX (quadratic) estimate of the proportion of treatment effect explained by the surrogate marker; only if estimator = "q"

`B1star.corrected.q`
the corrected SIMEX (quadratic) estimate of the adjusted regression coefficient for treatment; only if estimator = "q" and Ronly = FALSE and parametric = TRUE

`B1star.corrected.var.q`
the estimated variance of the corrected SIMEX (quadratic) estimate of the adjusted regression coefficient for treatment; only if estimator = "q" and Ronly = FALSE and parametric = TRUE

`B1star.corrected.CI.normal.q`
the 95% confidence interval using the normal approximation for the corrected SIMEX (quadratic) estimate of the adjusted regression coefficient for treatment; only if estimator = "q" and Ronly = FALSE and parametric = TRUE

`deltas.corrected.q`
the corrected SIMEX (quadratic) estimate of the residual treatment effect; only if estimator = "q" and Ronly = FALSE and parametric = FALSE

`deltas.corrected.var.q`
the estimated variance of the corrected SIMEX (quadratic) estimate of the residual treatment effect; only if estimator = "q" and Ronly = FALSE and parametric = FALSE

`deltas.corrected.CI.normal.q`
the 95% confidence interval using the normal approximation for the corrected SIMEX (quadratic) estimate of the residual treatment effect; only if estimator = "q" and Ronly = FALSE and parametric = FALSE

`R.corrected.nl` the corrected SIMEX (nonlinear) estimate of the proportion of treatment effect explained by the surrogate marker; only if estimator = "q"

`R.corrected.var.nl`
the estimated variance of the corrected SIMEX (nonlinear) estimate of the proportion of treatment effect explained by the surrogate marker; only if estimator = "q"

`R.corrected.CI.normal.nl`
the 95% confidence interval using the normal approximation for the corrected SIMEX (nonlinear) estimate of the proportion of treatment effect explained by the surrogate marker; only if estimator = "q"

`R.corrected.CI.fieller.nl`
the 95% confidence interval using Fieller's approach for the corrected SIMEX (nonlinear) estimate of the proportion of treatment effect explained by the surrogate marker; only if estimator = "q"

`B1star.corrected.nl`
the corrected SIMEX (nonlinear) estimate of the adjusted regression coefficient for treatment; only if estimator = "q" and Ronly = FALSE and parametric = TRUE

`B1star.corrected.var.nl`
the estimated variance of the corrected SIMEX (nonlinear) estimate of the adjusted regression coefficient for treatment; only if estimator = "q" and Ronly = FALSE and parametric = TRUE

`B1star.corrected.CI.normal.nl`
the 95% confidence interval using the normal approximation for the corrected SIMEX (nonlinear) estimate of the adjusted regression coefficient for treatment; only if estimator = "q" and Ronly = FALSE and parametric = TRUE

`deltas.corrected.nl`
the corrected SIMEX (nonlinear) estimate of the residual treatment effect; only if estimator = "q" and Ronly = FALSE and parametric = FALSE

`deltas.corrected.var.nl`
the estimated variance of the corrected SIMEX (nonlinear) estimate of the residual treatment effect; only if estimator = "q" and Ronly = FALSE and parametric = FALSE

`deltas.corrected.CI.normal.nl`
the 95% confidence interval using the normal approximation for the corrected SIMEX (nonlinear) estimate of the residual treatment effect; only if estimator = "q" and Ronly = FALSE and parametric = FALSE

Author(s)

Layla Parast

References

Parast, L., Garcia, T. P., Prentice, R. L., & Carroll, R. J. (2022). Robust methods to correct for measurement error when evaluating a surrogate marker. *Biometrics*, 78(1), 9-23.

Examples

```
data(d_example_me)
names(d_example_me)
R.s.estimate.me(yone=d_example_me$y1, yzero=d_example_me$y0, sone=d_example_me$s1,
szero=d_example_me$s0, parametric = TRUE, estimator = "d", me.variance = 0.5,
naive= TRUE, Ronly = FALSE)
R.s.estimate.me(yone=d_example_me$y1, yzero=d_example_me$y0, sone=d_example_me$s1,
szero=d_example_me$s0, parametric = TRUE, estimator = "q", me.variance = 0.5,
naive= FALSE, Ronly = TRUE)

#estimating measurement error variance with replicates
replicates = rbind(cbind(d_example_me$s1_rep1, d_example_me$s1_rep2,
d_example_me$s1_rep3), cbind(d_example_me$s0_rep1, d_example_me$s0_rep2,
d_example_me$s0_rep3))
mean.i = apply(replicates,1,mean, na.rm = TRUE)
num.i = apply(replicates,1,function(x) sum(!is.na(x)))
var.u = sum((replicates-mean.i)^2, na.rm = TRUE)/sum(num.i)
var.u
R.s.estimate.me(yone=d_example_me$y1, yzero=d_example_me$y0, sone=d_example_me$s1,
szero=d_example_me$s0, parametric = TRUE, estimator = "d", me.variance = var.u,
naive= TRUE, Ronly = FALSE)

R.s.estimate.me(yone=d_example_me$y1, yzero=d_example_me$y0,
sone=d_example_me$s1, szero=d_example_me$s0, parametric = FALSE, estimator = "q",
me.variance = 0.5, naive= FALSE, Ronly = TRUE)
```

R.s.surv.estimate	<i>Calculates the proportion of treatment effect explained by the surrogate marker information measured at a specified time and primary outcome information up to that specified time</i>
-------------------	---

Description

This function calculates the proportion of treatment effect on the primary outcome explained by the surrogate marker information measured at t_0 and primary outcome information up to t_0 . The user can also request a variance estimate, estimated using perturbing-resampling, and a 95% confidence interval. If a confidence interval is requested three versions are provided: a normal approximation based interval, a quantile based interval and Fieller's confidence interval, all using perturbation-resampling. The user can also request an estimate of the incremental value of surrogate marker information.

Usage

```
R.s.surv.estimate(xone, xzero, deltaone, deltazero, sone, szero, t,
weight.perturb = NULL, landmark, extrapolate = FALSE, transform = FALSE,
conf.int = FALSE, var = FALSE, incremental.value = FALSE, approx = T)
```

Arguments

xone	numeric vector, the observed event times in the treatment group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
xzero	numeric vector, the observed event times in the control group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
deltaone	numeric vector, the event indicators for the treatment group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
deltazero	numeric vector, the event indicators for the control group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
sone	numeric vector; surrogate marker measurement at t_0 for treated observations, assumed to be continuous. If $X_{1i} < t_0$, then the surrogate marker measurement should be NA.
szero	numeric vector; surrogate marker measurement at t_0 for control observations, assumed to be continuous. If $X_{1i} < t_0$, then the surrogate marker measurement should be NA.
t	the time of interest.
weight.perturb	weights used for perturbation resampling.
landmark	the landmark time t_0 or time of surrogate marker measurement.
extrapolate	TRUE or FALSE; indicates whether the user wants to use extrapolation.
transform	TRUE or FALSE; indicates whether the user wants to use a transformation for the surrogate marker.
conf.int	TRUE or FALSE; indicates whether a 95% confidence interval for delta is requested, default is FALSE.
var	TRUE or FALSE; indicates whether a variance estimate is requested, default is FALSE.

incremental.value

TRUE or FALSE; indicates whether the user would like to see the incremental value of the surrogate marker information, default is FALSE.

approx

TRUE or FALSE indicating whether an approximation should be used when calculating the probability of censoring; most relevant in settings where the survival time of interest for the primary outcome is greater than the last observed event but before the last censored case, default is TRUE.

Details

Let G be the binary treatment indicator with $G = 1$ for treatment and $G = 0$ for control and we assume throughout that subjects are randomly assigned to a treatment group at baseline. Let $T^{(1)}$ and $T^{(0)}$ denote the time of the primary outcome of interest, death for example, under the treatment and under the control, respectively. Let $S^{(1)}$ and $S^{(0)}$ denote the surrogate marker measured at time t_0 under the treatment and the control, respectively.

The residual treatment effect is defined as

$$\Delta_S(t, t_0) = P(T^{(0)} > t_0) \left\{ \int \psi_1(t|s, t_0) dF_0(s|t_0) - P(T^{(0)} > t|T^{(0)} > t_0) \right\}$$

where $F_0(\cdot|t_0)$ is the cumulative distribution function of $S^{(0)}$ conditional on $T^{(0)} > t_0$ and $\psi_1(t|s, t_0) = P(T^{(1)} > t|S^{(1)} = s, T^{(1)} > t_0)$. The proportion of treatment effect explained by the surrogate marker information measured at t_0 and primary outcome information up to t_0 , which we denote by $R_S(t, t_0)$, can be expressed using a contrast between $\Delta_S(t, t_0)$ and $\Delta(t)$:

$$R_S(t, t_0) = \{\Delta(t) - \Delta_S(t, t_0)\} / \Delta(t) = 1 - \Delta_S(t, t_0) / \Delta(t).$$

The definition and estimation of $\Delta(t)$ is described in the delta.surv.estimate documentation.

Due to censoring, our data consist of n_1 observations $\{(X_{1i}, \delta_{1i}, S_{1i}), i = 1, \dots, n_1\}$ from the treatment group $G = 1$ and n_0 observations $\{(X_{0i}, \delta_{0i}, S_{0i}), i = 1, \dots, n_0\}$ from the control group $G = 0$ where $X_{gi} = \min(T_{gi}, C_{gi})$, $\delta_{gi} = I(T_{gi} < C_{gi})$, C_{gi} denotes the censoring time, and S_{gi} denotes the surrogate marker information measured at time t_0 , for $g = 1, 0$, for individual i . Note that if $X_{gi} < t_0$, then S_{gi} should be NA (not available).

To estimate $\Delta_S(t, t_0)$, we use a nonparametric kernel Nelson-Aalen estimator to estimate $\psi_1(t|s, t_0)$ as $\hat{\psi}_1(t|s, t_0) = \exp\{-\hat{\Lambda}_1(t|s, t_0)\}$, where

$$\hat{\Lambda}_1(t|s, t_0) = \int_{t_0}^t \frac{\sum_{i=1}^{n_1} I(X_{1i} > t_0) K_h\{\gamma(S_{1i}) - \gamma(s)\} dN_{1i}(z)}{\sum_{i=1}^{n_1} I(X_{1i} > t_0) K_h\{\gamma(S_{1i}) - \gamma(s)\} Y_{1i}(z)},$$

is a consistent estimate of $\Lambda_1(t|s, t_0) = -\log[\psi_1(t|s, t_0)]$, $Y_{1i}(t) = I(X_{1i} \geq t)$, $N_{1i}(t) = I(X_{1i} \leq t)\delta_{1i}$, $K(\cdot)$ is a smooth symmetric density function, $K_h(x) = K(x/h)/h$, $\gamma(\cdot)$ is a given monotone transformation function, and h is a specified bandwidth. To obtain an appropriate h we first use `bw.nrd` to obtain h_{opt} ; and then we let $h = h_{opt}n_1^{-c_0}$ with $c_0 = 0.11$.

Since $F_0(s|t_0) = P(S_{0i} \leq s|X_{0i} > t_0)$, we empirically estimate $F_0(s|t_0)$ using all subjects with $X_{0i} > t_0$ as

$$\hat{F}_0(s|t_0) = \frac{\sum_{i=1}^{n_0} I(S_{0i} \leq s, X_{0i} > t_0)}{\sum_{i=1}^{n_0} I(X_{0i} > t_0)}.$$

Subsequently, we construct an estimator for $\Delta_S(t, t_0)$ as

$$\hat{\Delta}_S(t, t_0) = n_0^{-1} \sum_{i=1}^{n_0} \left[\hat{\psi}_1(t|S_{0i}, t_0) \frac{I(X_{0i} > t_0)}{\hat{W}_0^C(t_0)} - \frac{I(X_{0i} > t)}{\hat{W}_0^C(t)} \right]$$

where $\hat{W}_g^C(\cdot)$ is the Kaplan-Meier estimator of survival for censoring for $g = 1, 0$. Finally, we estimate $R_S(t, t_0)$ as $\hat{R}_S(t, t_0) = 1 - \hat{\Delta}_S(t, t_0)/\hat{\Delta}(t)$.

Variance estimation and confidence interval construction are performed using perturbation-resampling. Specifically, let $\{V^{(b)} = (V_{11}^{(b)}, \dots, V_{1n_1}^{(b)}, V_{01}^{(b)}, \dots, V_{0n_0}^{(b)})^T, b = 1, \dots, D\}$ be $n \times D$ independent copies of a positive random variables V from a known distribution with unit mean and unit variance. Let

$$\hat{\Delta}^{(b)}(t) = \frac{\sum_{i=1}^{n_1} V_{1i}^{(b)} I(X_{1i} > t)}{\sum_{i=1}^{n_1} V_{1i}^{(b)} \hat{W}_1^{C(b)}(t)} - \frac{\sum_{i=1}^{n_0} V_{0i}^{(b)} I(X_{0i} > t)}{\sum_{i=1}^{n_0} V_{0i}^{(b)} \hat{W}_0^{C(b)}(t)}.$$

In this package, we use weights generated from an Exponential(1) distribution and use $D = 500$. The variance of $\hat{\Delta}(t)$ is obtained as the empirical variance of $\{\hat{\Delta}(t)^{(b)}, b = 1, \dots, D\}$. Variance estimates for $\hat{\Delta}_S(t, t_0)$ and $\hat{R}_S(t, t_0)$ are calculated similarly. We construct two versions of the 95% confidence interval for each estimate: one based on a normal approximation confidence interval using the estimated variance and another taking the 2.5th and 97.5th empirical percentile of the perturbed quantities. In addition, we use Fieller's method to obtain a third confidence interval for $R_S(t, t_0)$ as

$$\left\{ 1 - r : \frac{(\hat{\Delta}_S(t, t_0) - r\hat{\Delta}(t))^2}{\hat{\sigma}_{11} - 2r\hat{\sigma}_{12} + r^2\hat{\sigma}_{22}} \leq c_\alpha \right\},$$

where $\hat{\Sigma} = (\hat{\sigma}_{ij})_{1 \leq i, j \leq 2}$ and c_α is the $(1 - \alpha)$ th percentile of

$$\left\{ \frac{\{\hat{\Delta}_S^{(b)}(t) - (1 - \hat{R}_S(t, t_0))\hat{\Delta}(t)^{(b)}\}^2}{\hat{\sigma}_{11} - 2(1 - \hat{R}_S(t, t_0))\hat{\sigma}_{12} + (1 - \hat{R}_S(t, t_0))^2\hat{\sigma}_{22}}, b = 1, \dots, D \right\}$$

where $\alpha = 0.05$.

Since the definition of $R_S(t, t_0)$ considers the surrogate information as a combination of both S information and T information up to t_0 , a logical inquiry would be how to assess the incremental value of the S information in terms of the proportion of treatment effect explained, when added to T information up to t_0 . The proportion of treatment effect explained by T information up to t_0 only is denoted as $R_T(t, t_0)$ and is described in the documentation for R.t.surv.estimate. The incremental value of S information is defined as:

$$IV_S(t, t_0) = R_S(t, t_0) - R_T(t, t_0) = \frac{\Delta_T(t, t_0) - \Delta_S(t, t_0)}{\Delta(t)}.$$

For estimation of $R_T(t_0)$, see documentation for R.t.surv.estimate. The quantity $IV_S(t, t_0)$ is then estimated by $\hat{IV}_S(t, t_0) = \hat{R}_S(t, t_0) - \hat{R}_T(t, t_0)$. Perturbation-resampling is used for variance estimation and confidence interval construction for this quantity, similar to the other quantities in this package.

Note that if the observed supports for S are not the same, then $\hat{\Lambda}_1(t|s, t_0)$ for $S_{0i} = s$ outside the support of S_{1i} may return NA (depending on the bandwidth). If `extrapolation = TRUE`, then the $\hat{\Lambda}_1(t|s, t_0)$ values for these surrogate values are set to the closest non-NA value. If `transform = TRUE`, then S_{1i} and S_{0i} are transformed such that the new transformed values, S_{1i}^{tr} and S_{0i}^{tr} are defined as: $S_{gi}^{tr} = F([S_{gi} - \mu]/\sigma)$ for $g = 0, 1$ where $F(\cdot)$ is the cumulative distribution function for a standard normal random variable, and μ and σ are the sample mean and standard deviation, respectively, of $\{(S_{1i}, S_{0i})^T, i \text{ s.t. } X_{gi} > t_0\}$.

Value

A list is returned:

`delta` the estimate, $\hat{\Delta}(t)$, described in `delta.estimate` documentation.

<code>delta.s</code>	the estimate, $\hat{\Delta}_S(t, t_0)$, described above.
<code>R.s</code>	the estimate, $\hat{R}_S(t, t_0)$, described above.
<code>delta.var</code>	the variance estimate of $\hat{\Delta}(t)$; if <code>var = TRUE</code> or <code>conf.int = TRUE</code> .
<code>delta.s.var</code>	the variance estimate of $\hat{\Delta}_S(t, t_0)$; if <code>var = TRUE</code> or <code>conf.int = TRUE</code> .
<code>R.s.var</code>	the variance estimate of $\hat{R}_S(t, t_0)$; if <code>var = TRUE</code> or <code>conf.int = TRUE</code> .
<code>conf.int.normal.delta</code>	a vector of size 2; the 95% confidence interval for $\hat{\Delta}(t)$ based on a normal approximation; if <code>conf.int = TRUE</code> .
<code>conf.int.quantile.delta</code>	a vector of size 2; the 95% confidence interval for $\hat{\Delta}(t)$ based on sample quantiles of the perturbed values, described above; if <code>conf.int = TRUE</code> .
<code>conf.int.normal.delta.s</code>	a vector of size 2; the 95% confidence interval for $\hat{\Delta}_S(t, t_0)$ based on a normal approximation; if <code>conf.int = TRUE</code> .
<code>conf.int.quantile.delta.s</code>	a vector of size 2; the 95% confidence interval for $\hat{\Delta}_S(t, t_0)$ based on sample quantiles of the perturbed values, described above; if <code>conf.int = TRUE</code> .
<code>conf.int.normal.R.s</code>	a vector of size 2; the 95% confidence interval for $\hat{R}_S(t, t_0)$ based on a normal approximation; if <code>conf.int = TRUE</code> .
<code>conf.int.quantile.R.s</code>	a vector of size 2; the 95% confidence interval for $\hat{R}_S(t, t_0)$ based on sample quantiles of the perturbed values, described above; if <code>conf.int = TRUE</code> .
<code>conf.int.fieller.R.s</code>	a vector of size 2; the 95% confidence interval for $\hat{R}_S(t, t_0)$ based on Fieller's approach, described above; if <code>conf.int = TRUE</code> .
<code>delta.t</code>	the estimate, $\hat{\Delta}_T(t, t_0)$, described above; if <code>incremental.vaue = TRUE</code> .
<code>R.t</code>	the estimate, $\hat{R}_T(t, t_0)$, described above; if <code>incremental.vaue = TRUE</code> .
<code>incremental.value</code>	the estimate, $\hat{IV}_S(t, t_0)$, described above; if <code>incremental.vaue = TRUE</code> .
<code>delta.t.var</code>	the variance estimate of $\hat{\Delta}_T(t, t_0)$; if <code>var = TRUE</code> or <code>conf.int = TRUE</code> and <code>incremental.vaue = TRUE</code> .
<code>R.t.var</code>	the variance estimate of $\hat{R}_T(t, t_0)$; if <code>var = TRUE</code> or <code>conf.int = TRUE</code> and <code>incremental.vaue = TRUE</code> .
<code>incremental.value.var</code>	the variance estimate of $\hat{IV}_S(t, t_0)$; if <code>var = TRUE</code> or <code>conf.int = TRUE</code> and <code>incremental.vaue = TRUE</code> .
<code>conf.int.normal.delta.t</code>	a vector of size 2; the 95% confidence interval for $\hat{\Delta}_T(t, t_0)$ based on a normal approximation; if <code>conf.int = TRUE</code> and <code>incremental.vaue = TRUE</code> .
<code>conf.int.quantile.delta.t</code>	a vector of size 2; the 95% confidence interval for $\hat{\Delta}_T(t, t_0)$ based on sample quantiles of the perturbed values, described above; if <code>conf.int = TRUE</code> and <code>incremental.vaue = TRUE</code> .
<code>conf.int.normal.R.t</code>	a vector of size 2; the 95% confidence interval for $\hat{R}_T(t, t_0)$ based on a normal approximation; if <code>conf.int = TRUE</code> and <code>incremental.vaue = TRUE</code> .

`conf.int.quantile.R.t`
 a vector of size 2; the 95% confidence interval for $\hat{R}_T(t, t_0)$ based on sample quantiles of the perturbed values, described above; if `conf.int = TRUE` and `incremental.vaue = TRUE`.

`conf.int.fieller.R.t`
 a vector of size 2; the 95% confidence interval for $\hat{R}_T(t, t_0)$ based on Fieller's approach, described above; if `conf.int = TRUE` and `incremental.vaue = TRUE`.

`conf.int.normal.iv`
 a vector of size 2; the 95% confidence interval for $\hat{IV}_S(t, t_0)$ based on a normal approximation; if `conf.int = TRUE` and `incremental.vaue = TRUE`.

`conf.int.quantile.iv`
 a vector of size 2; the 95% confidence interval for $\hat{IV}_S(t, t_0)$ based on sample quantiles of the perturbed values, described above; if `conf.int = TRUE` and `incremental.vaue = TRUE`.

Note

If the treatment effect is not significant, the user will receive the following message: "Warning: it looks like the treatment effect is not significant; may be difficult to interpret the residual treatment effect in this setting". If the observed support of the surrogate marker for the control group is outside the observed support of the surrogate marker for the treatment group, the user will receive the following message: "Warning: observed supports do not appear equal, may need to consider a transformation or extrapolation".

Author(s)

Layla Parast

References

Parast, L., Cai, T., & Tian, L. (2017). Evaluating surrogate marker information using censored data. *Statistics in Medicine*, 36(11), 1767-1782.

Examples

```
data(d_example_surv)
names(d_example_surv)
```

<code>R.t.surv.estimate</code>	<i>Calculates the proportion of treatment effect explained by the primary outcome information up to a specified time</i>
--------------------------------	--

Description

This function calculates the proportion of treatment effect on the primary outcome explained by the treatment effect on the primary outcome up to t_0 . The user can also request a variance estimate, estimated using perturbing-resampling, and a 95% confidence interval. If a confidence interval is requested three versions are provided: a normal approximation based interval, a quantile based interval and Fieller's confidence interval, all using perturbation-resampling.

Usage

```
R.t.surv.estimate(xone, xzero, deltaone, deltazero, t, weight.perturb = NULL,
landmark, var = FALSE, conf.int = FALSE, approx = T)
```

Arguments

xone	numeric vector, the observed event times in the treatment group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
xzero	numeric vector, the observed event times in the control group, $X = \min(T, C)$ where T is the time of the primary outcome and C is the censoring time.
deltaone	numeric vector, the event indicators for the treatment group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
deltazero	numeric vector, the event indicators for the control group, $D = I(T < C)$ where T is the time of the primary outcome and C is the censoring time.
t	the time of interest.
weight.perturb	weights used for perturbation resampling.
landmark	the landmark time t_0 or time of surrogate marker measurement.
var	TRUE or FALSE; indicates whether a variance estimate for delta is requested, default is FALSE.
conf.int	TRUE or FALSE; indicates whether a 95% confidence interval for delta is requested, default is FALSE.
approx	TRUE or FALSE indicating whether an approximation should be used when calculating the probability of censoring; most relevant in settings where the survival time of interest for the primary outcome is greater than the last observed event but before the last censored case, default is TRUE.

Details

Let G be the binary treatment indicator with $G = 1$ for treatment and $G = 0$ for control and we assume throughout that subjects are randomly assigned to a treatment group at baseline. Let T denote the time of the primary outcome of interest, death for example. We use potential outcomes notation such that $T^{(g)}$ denotes the time of the primary outcome under treatment $G = g$. The proportion of treatment effect explained by T observed up to t_0 only is $R_T(t, t_0) = 1 - \Delta_T(t, t_0)/\hat{\Delta}(t)$ where

$$\Delta_T(t, t_0) = P(T^{(0)} > t_0)P(T^{(1)} > t \mid T^{(1)} > t_0) - P(T^{(0)} > t).$$

To estimate $R_T(t, t_0)$, we use the estimator $\hat{R}_T(t, t_0) = 1 - \hat{\Delta}_T(t, t_0)/\hat{\Delta}(t)$ where $\hat{\Delta}_T(t, t_0) = \hat{\phi}_0(t_0)\hat{\phi}_1(t)/\hat{\phi}_1(t_0) - \hat{\phi}_0(t)$ and $\hat{\phi}_g(u) = n_g^{-1} \sum_{i=1}^{n_g} \frac{I(X_{gi} > u)}{\hat{W}_g^C(u)}$ for $g = 1, 0$ where $\hat{W}_g^C(\cdot)$ is the Kaplan-Meier estimator of survival for censoring for $g = 1, 0$.

Value

A list is returned:

delta	the estimate, $\hat{\Delta}(t)$, described in delta.estimate documentation.
delta.t	the estimate, $\hat{\Delta}_T(t, t_0)$, described above.
R.t	the estimate, $\hat{R}_T(t, t_0)$, described above.
delta.var	the variance estimate of $\hat{\Delta}(t)$; if var = TRUE or conf.int = TRUE.
delta.t.var	the variance estimate of $\hat{\Delta}_T(t, t_0)$; if var = TRUE or conf.int = TRUE.

`R.t.var` the variance estimate of $\hat{R}_T(t, t_0)$; if `var = TRUE` or `conf.int = TRUE`.

`conf.int.normal.delta` a vector of size 2; the 95% confidence interval for $\hat{\Delta}(t)$ based on a normal approximation; if `conf.int = TRUE`.

`conf.int.quantile.delta` a vector of size 2; the 95% confidence interval for $\hat{\Delta}(t)$ based on sample quantiles of the perturbed values, described above; if `conf.int = TRUE`.

`conf.int.normal.delta.t` a vector of size 2; the 95% confidence interval for $\hat{\Delta}_T(t, t_0)$ based on a normal approximation; if `conf.int = TRUE`.

`conf.int.quantile.delta.t` a vector of size 2; the 95% confidence interval for $\hat{\Delta}_T(t, t_0)$ based on sample quantiles of the perturbed values, described above; if `conf.int = TRUE`.

`conf.int.normal.R.t` a vector of size 2; the 95% confidence interval for $\hat{R}_T(t, t_0)$ based on a normal approximation; if `conf.int = TRUE`.

`conf.int.quantile.R.t` a vector of size 2; the 95% confidence interval for $\hat{R}_T(t, t_0)$ based on sample quantiles of the perturbed values, described above; if `conf.int = TRUE`.

`conf.int.fieller.R.t` a vector of size 2; the 95% confidence interval for $\hat{R}_T(t, t_0)$ based on Fieller's approach, described above; if `conf.int = TRUE`.

Note

If the treatment effect is not significant, the user will receive the following message: "Warning: it looks like the treatment effect is not significant; may be difficult to interpret the residual treatment effect in this setting".

Author(s)

Layla Parast

References

Parast, L., Cai, T., & Tian, L. (2017). Evaluating surrogate marker information using censored data. *Statistics in Medicine*, 36(11), 1767-1782.

Examples

```
data(d_example_surv)
names(d_example_surv)
```

select.surrogate	<i>Selects an optimal subset of surrogate markers and estimates the proportion of treatment effect explained</i>
------------------	--

Description

This function estimates the proportion of treatment effect on the primary outcome explained by the treatment effect on a set of surrogate markers, optionally first selecting a subset of the candidate markers using treatment-aware regularization (TAR) or an unweighted regularized approach. The user can also request a bootstrap variance estimate and 95% confidence intervals.

Usage

```
select.surrogate(sone, szero, yone, yzero, selection = "none", var = FALSE,
boot.num = 200, nlambda = 20, folds = 5, epsilon = 0.05, c.constant = 1,
small.n = 1e-6, lambda_rule = "lambda.1se")
```

Arguments

sone	numeric matrix or data frame; candidate surrogate markers for treated observations, with n_1 (number of treated observations) rows and k (number of candidate surrogate markers) columns. If the columns are not named, they are labeled s1, ..., sk.
szero	numeric matrix or data frame; candidate surrogate markers for control observations, with n_0 (number of control observations) rows and k columns, in the same order as sone.
yone	numeric vector; primary outcome for treated observations.
yzero	numeric vector; primary outcome for control observations.
selection	specifies how the subset of surrogate markers is chosen; choices are "none" (no selection; the proportion of treatment effect explained is estimated using all candidate markers, as in R.s.estimate with number = "multiple"), "TAR" (treatment-aware regularization), or "rr" (an unweighted, regularized approach), default is "none".
var	TRUE or FALSE; indicates whether a bootstrap variance estimate and 95% confidence intervals are requested, default is FALSE.
boot.num	number of bootstrap resamples used when var = TRUE; on each resample, both treatment groups are resampled with replacement and the entire selection procedure, including re-selection of the penalty, is rerun, default is 200.
nlambda	the number of penalty values considered along the lasso path used by selection = "TAR" or selection = "rr", default is 20.
folds	the number of cross-validation folds, within the treated group, used by selection = "TAR" to select the penalty, default is 5. Not used if selection is "none" or "rr".
epsilon	the tolerance used by selection = "TAR": the selected subset is the sparsest point along the weighted lasso path whose estimated proportion of treatment effect explained falls no more than epsilon below that of the full candidate set, default is 0.05. Not used if selection is "none" or "rr".

c.constant	a constant controlling the strength of the treatment-aware penalty weighting in selection = "TAR", default is 1. Not used if selection is "none" or "rr".
small.n	a small positive constant added to the estimated treatment effect magnitude of each candidate marker before inverting it to form the treatment-aware penalty weights in selection = "TAR", to avoid dividing by a value at or near zero, default is 1e-6. Not used if selection is "none" or "rr".
lambda_rule	the cross-validation rule used to select the penalty in selection = "rr"; choices are "lambda.1se" or "lambda.min", default is "lambda.1se". Not used if selection is "none" or "TAR".

Details

Let $Y^{(1)}$ and $Y^{(0)}$ denote the primary outcome under treatment and control, respectively, and let $\mathbf{S}^{(1)}$ and $\mathbf{S}^{(0)}$ denote the vector of k candidate surrogate markers under treatment and control, respectively. As in R.s.estimate, the proportion of treatment effect explained by a set of markers $\mathbf{S}$ is $R_S = 1 - \Delta_S/\Delta$, where Δ is the overall treatment effect and Δ_S is the residual treatment effect after accounting for $\mathbf{S}$, estimated using a working model to reduce the dimension of $\mathbf{S}$ followed by kernel smoothing, exactly as described in the R.s.estimate documentation for number = "multiple", type = "robust".

When selection = "none", all k candidate markers are used, and the estimate of R_S agrees with that of R.s.estimate run with number = "multiple".

When selection = "TAR" or selection = "rr", a subset $\tilde{\mathbf{S}}$ of the k candidate markers is first selected via a lasso fit to the treated group, and $R_{\tilde{\mathbf{S}}}$ is then estimated for the selected subset only, using a debiased (unpenalized) refit on the selected markers followed by the same kernel-smoothed approach. For selection = "rr", the lasso is unweighted and the penalty is chosen by ordinary cross-validation using lambda_rule. For selection = "TAR", each candidate marker's lasso penalty is instead weighted by the inverse of its estimated standardized treatment effect, so that markers with little or no treatment effect are penalized more heavily, and the penalty along the resulting weighted lasso path is selected via cross-validation, within the treated group, to be the sparsest point whose estimated proportion of treatment effect explained falls no more than epsilon below that of the full candidate set.

If a bootstrap variance estimate and confidence intervals are requested (var = TRUE), both treatment groups are resampled with replacement and the entire estimation procedure specified by selection is rerun on each resample, including re-selection of the penalty for selection = "TAR" or selection = "rr". This differs from the perturbation-resampling used elsewhere in this package. A 95% confidence interval is provided for $\hat{\Delta}$, $\hat{\Delta}_S$, and $\hat{R}_S$ (or $\hat{R}_{\tilde{\mathbf{S}}}$), based on the 2.5th and 97.5th percentiles of the bootstrap replicates.

If the observed supports for the candidate markers (or the selected subset) are not the same across treatment groups, values outside the observed support are extrapolated to the closest non-missing value; unlike R.s.estimate, this is done automatically and cannot be turned off.

Value

A list is returned:

delta	the estimate, $\hat{\Delta}$, described above.
delta.s	the estimate, $\hat{\Delta}_S$ (or $\hat{\Delta}_{\tilde{\mathbf{S}}}$ if selection is "TAR" or "rr"), described above.
R.s	the estimate, $\hat{R}_S$ (or $\hat{R}_{\tilde{\mathbf{S}}}$ if selection is "TAR" or "rr"), described above.
selection	a named numeric vector of the (penalized) lasso coefficients of the selected markers; only returned if selection is "TAR" or "rr".

delta.var	the bootstrap variance estimate of $\hat{\Delta}$; if var = TRUE.
delta.s.var	the bootstrap variance estimate of $\hat{\Delta}_S$; if var = TRUE.
R.s.var	the bootstrap variance estimate of $\hat{R}_S$; if var = TRUE.
conf.int.delta	a vector of size 2; the 95% confidence interval for $\hat{\Delta}$, based on the 2.5th and 97.5th percentiles of the bootstrap replicates; if var = TRUE.
conf.int.delta.s	a vector of size 2; the 95% confidence interval for $\hat{\Delta}_S$, based on the 2.5th and 97.5th percentiles of the bootstrap replicates; if var = TRUE.
conf.int.R.s	a vector of size 2; the 95% confidence interval for $\hat{R}_S$, based on the 2.5th and 97.5th percentiles of the bootstrap replicates; if var = TRUE.

Author(s)

Layla Parast

References

Parast, L., Freestone, G., Tarr, J. (2026). Treatment-Aware Regularization via Multi-Objective Optimization for Surrogate Selection.

Parast, L., McDermott, M., Tian, L. (2016). Robust estimation of the proportion of treatment effect explained by surrogate marker information. *Statistics in Medicine*, 35(10):1637-1653.

Examples

```
data(d_tar_example)
names(d_tar_example)
select.surrogate(sone = d_tar_example$sone, szero = d_tar_example$szero,
yone = d_tar_example$yone, yzero = d_tar_example$yzero, selection = "none")
select.surrogate(sone = d_tar_example$sone, szero = d_tar_example$szero,
yone = d_tar_example$yone, yzero = d_tar_example$yzero, selection = "TAR")
## Not run:
select.surrogate(sone = d_tar_example$sone, szero = d_tar_example$szero,
yone = d_tar_example$yone, yzero = d_tar_example$yzero, selection = "TAR",
var = TRUE, boot.num = 200)

## End(Not run)
```

VTM	<i>Repeats a row.</i>
-----	-----------------------

Description

Helper function; this function creates a matrix that repeats vc, dm times where each row is equal to the vc vector.

Usage

```
VTM(vc, dm)
```

Arguments

vc	the vector to repeat.
dm	number of rows.

Value

a matrix that repeats vc, dm times where each row is equal to the vc vector

Index

- * **arith**
 - cumsum2, 17
 - delper.si, 18
- * **array**
 - VTM, 62
- * **augment**
 - Aug.R.s.surv.estimate, 11
- * **datasets**
 - d_example, 28
 - d_example_me, 29
 - d_example_multiple, 30
 - d_example_surv, 30
 - d_tar_example, 31
- * **htest**
 - calculate.var.np, 15
 - fieller.ci, 33
 - me.variance.estimate, 34
- * **internal**
 - .bootstrap_var, 2
 - .debias_deltas, 3
 - .debias_ols, 4
 - .debias_p, 5
 - .debias_pred, 5
 - .full.surrogate.approach, 6
 - .make.delta.s, 7
 - .smoothed.regularized, 8
 - .tar.approach, 9
 - .te.calculate, 10
 - augment.est.vector, 15
 - calculate.var.np, 15
 - censor.weight, 16
 - cumsum2, 17
 - delper.si, 18
 - fieller.calculate.me, 32
 - Kern.FUN, 34
 - perturb.nu.vector, 35
 - pred.smooth, 35
 - pred.smooth.surv, 36
 - VTM, 62
- * **models**
 - .debias_ols, 4
 - delta.s.estimate, 22
 - R.s.estimate, 42
 - R.s.estimate.me, 46
 - select.surrogate, 60
- * **nonparametric**
 - .full.surrogate.approach, 6
 - .make.delta.s, 7
 - Aug.R.s.surv.estimate, 11
 - censor.weight, 16
 - delta.multiple.surv, 20
 - delta.s.estimate, 22
 - delta.s.surv.estimate, 23
 - delta.surv.estimate, 25
 - delta.t.surv.estimate, 27
 - Kern.FUN, 34
 - pred.smooth, 35
 - pred.smooth.surv, 36
 - R.multiple.surv, 37
 - R.s.estimate, 42
 - R.s.estimate.me, 46
 - R.s.surv.estimate, 53
 - R.t.surv.estimate, 57
 - select.surrogate, 60
- * **regression**
 - .debias_deltas, 3
 - .debias_ols, 4
 - .debias_p, 5
 - .debias_pred, 5
 - .full.surrogate.approach, 6
 - .make.delta.s, 7
 - .smoothed.regularized, 8
 - .tar.approach, 9
 - .te.calculate, 10
 - delta.s.estimate, 22
 - R.s.estimate, 42
 - R.s.estimate.me, 46
 - select.surrogate, 60
- * **robust**
 - .bootstrap_var, 2
 - .debias_deltas, 3
 - .debias_ols, 4
 - .debias_pred, 5
 - .full.surrogate.approach, 6
 - .make.delta.s, 7
 - .smoothed.regularized, 8

- .tar.approach, 9
- .te.calculate, 10
- Aug.R.s.surv.estimate, 11
- delta.multiple.surv, 20
- delta.s.estimate, 22
- delta.s.surv.estimate, 23
- delta.t.surv.estimate, 27
- Kern.FUN, 34
- pred.smooth, 35
- pred.smooth.surv, 36
- R.multiple.surv, 37
- R.s.estimate, 42
- R.s.estimate.me, 46
- R.s.surv.estimate, 53
- R.t.surv.estimate, 57
- select.surrogate, 60
- * smooth**
 - .debias_deltas, 3
 - .full.surrogate.approach, 6
 - .make.delta.s, 7
 - .smoothed.regularized, 8
 - .tar.approach, 9
 - Aug.R.s.surv.estimate, 11
 - delta.multiple.surv, 20
 - delta.s.estimate, 22
 - delta.s.surv.estimate, 23
 - Kern.FUN, 34
 - pred.smooth, 35
 - pred.smooth.surv, 36
 - R.multiple.surv, 37
 - R.s.estimate, 42
 - R.s.estimate.me, 46
 - R.s.surv.estimate, 53
 - select.surrogate, 60
- * survival**
 - Aug.R.s.surv.estimate, 11
 - censor.weight, 16
 - delta.multiple.surv, 20
 - delta.s.surv.estimate, 23
 - delta.surv.estimate, 25
 - delta.t.surv.estimate, 27
 - pred.smooth.surv, 36
 - R.multiple.surv, 37
 - R.s.surv.estimate, 53
 - R.t.surv.estimate, 57
- * univar**
 - delta.estimate, 18
 - delta.surv.estimate, 25
 - me.variance.estimate, 34
- .bootstrap_var, 2
- .debias_deltas, 3
- .debias_ols, 4
- .debias_p, 5
- .debias_pred, 5
- .full.surrogate.approach, 6
- .make.delta.s, 7
- .smoothed.regularized, 8
- .tar.approach, 9
- .te.calculate, 10
- Aug.R.s.surv.estimate, 11
- augment.est.vector, 15
- calculate.var.np, 15
- censor.weight, 16
- cumsum2, 17
- d_example, 28
- d_example_me, 29
- d_example_multiple, 30
- d_example_surv, 30
- d_tar_example, 31
- delper.si, 18
- delta.estimate, 18
- delta.multiple.surv, 20
- delta.s.estimate, 22
- delta.s.surv.estimate, 23
- delta.surv.estimate, 25
- delta.t.surv.estimate, 27
- fieller.calculate.me, 32
- fieller.ci, 33
- helper.si (delper.si), 18
- Kern.FUN, 34
- me.variance.estimate, 34
- perturb.nu.vector, 35
- pred.smooth, 35
- pred.smooth.surv, 36
- R.multiple.surv, 37
- R.s.estimate, 42
- R.s.estimate.me, 46
- R.s.surv.estimate, 53
- R.t.surv.estimate, 57
- select.surrogate, 60
- VTM, 62